

PG142-B00
DP + DP + DP + HDMI/DP

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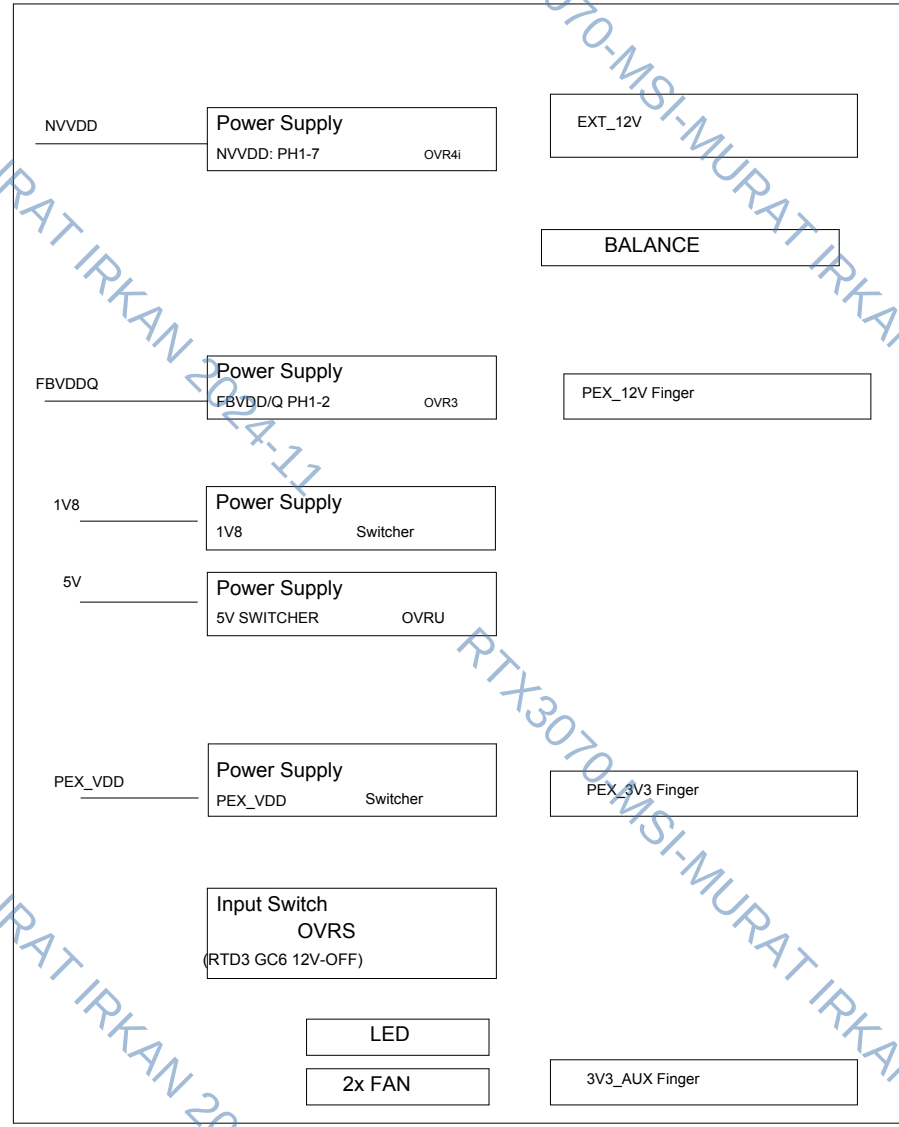
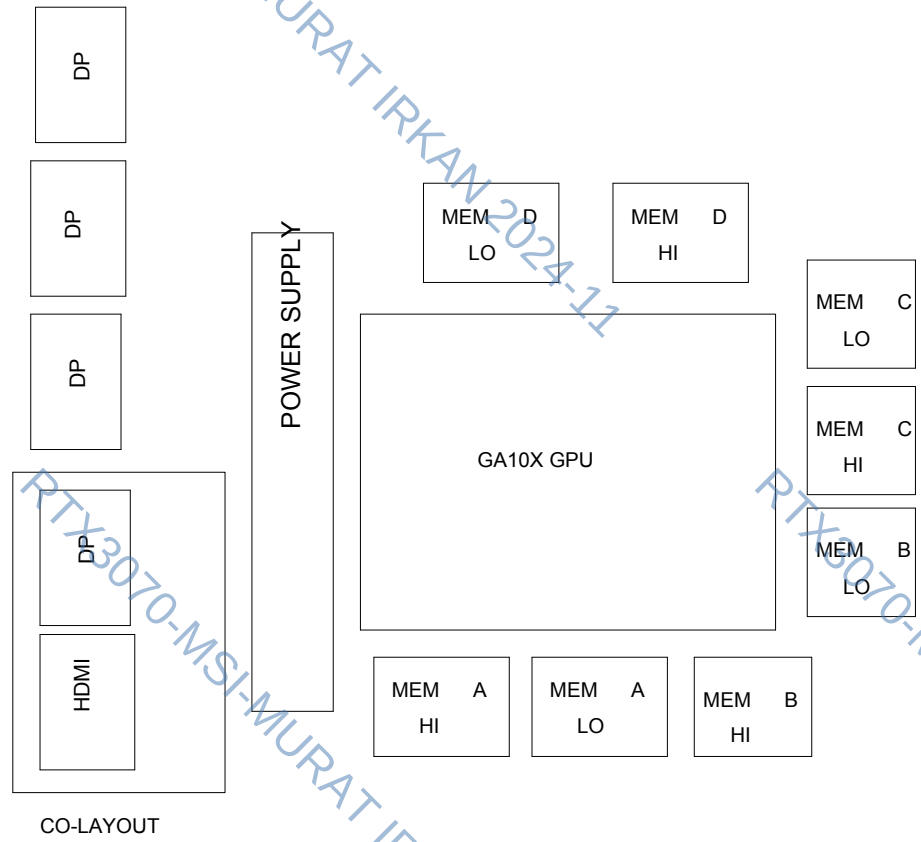
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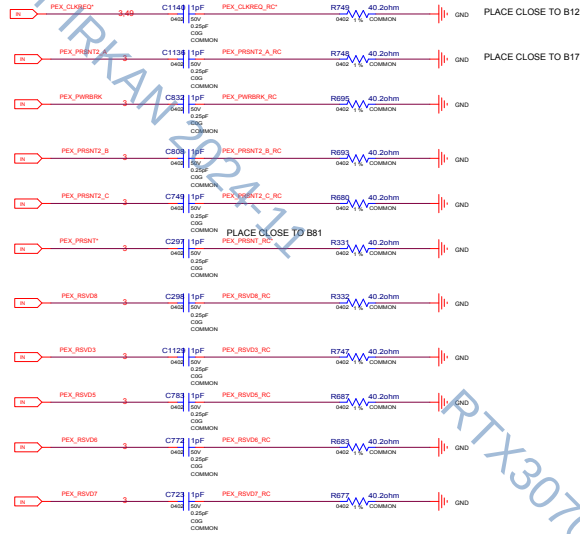
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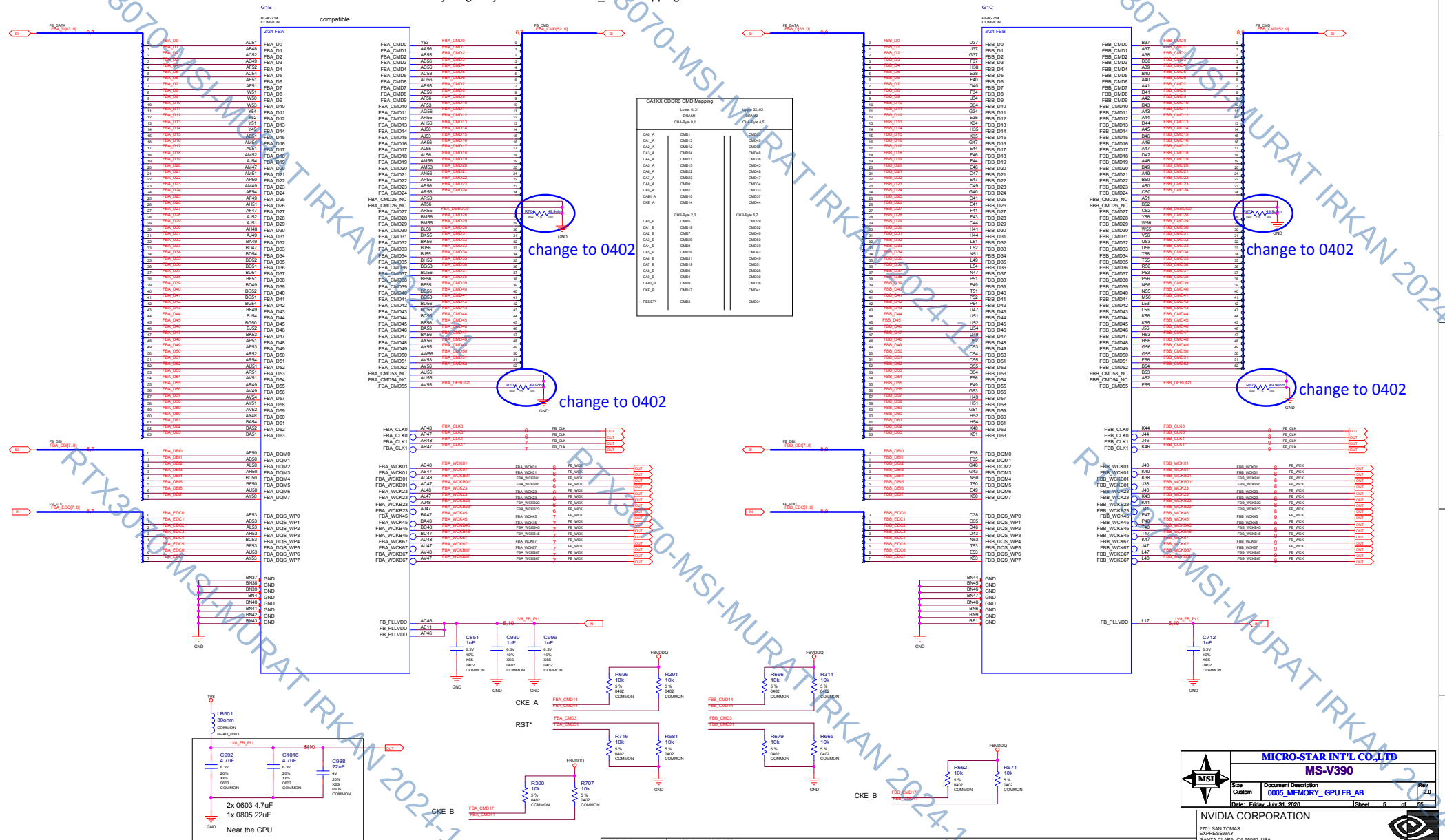
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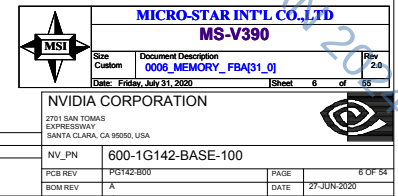
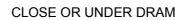
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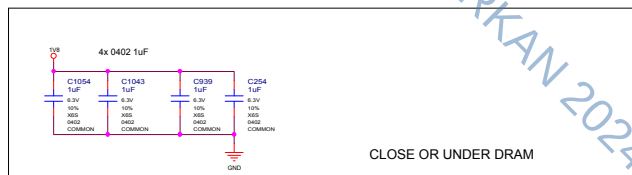
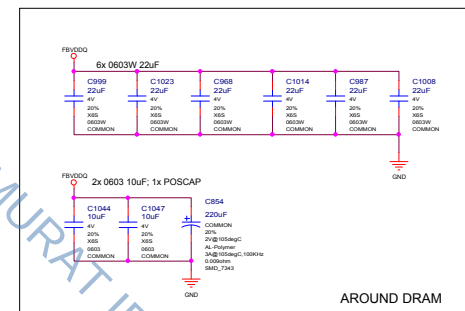
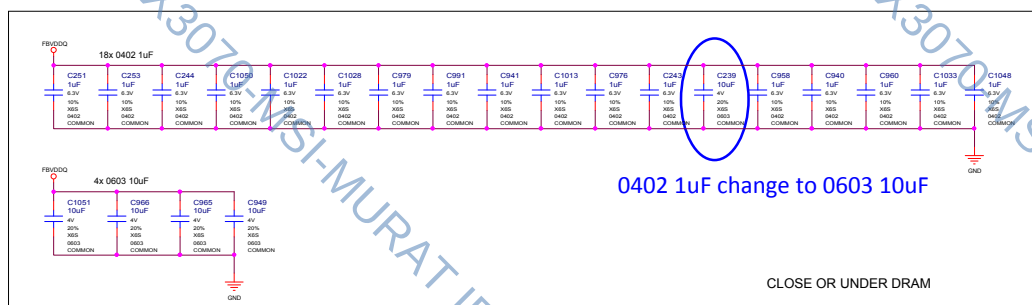
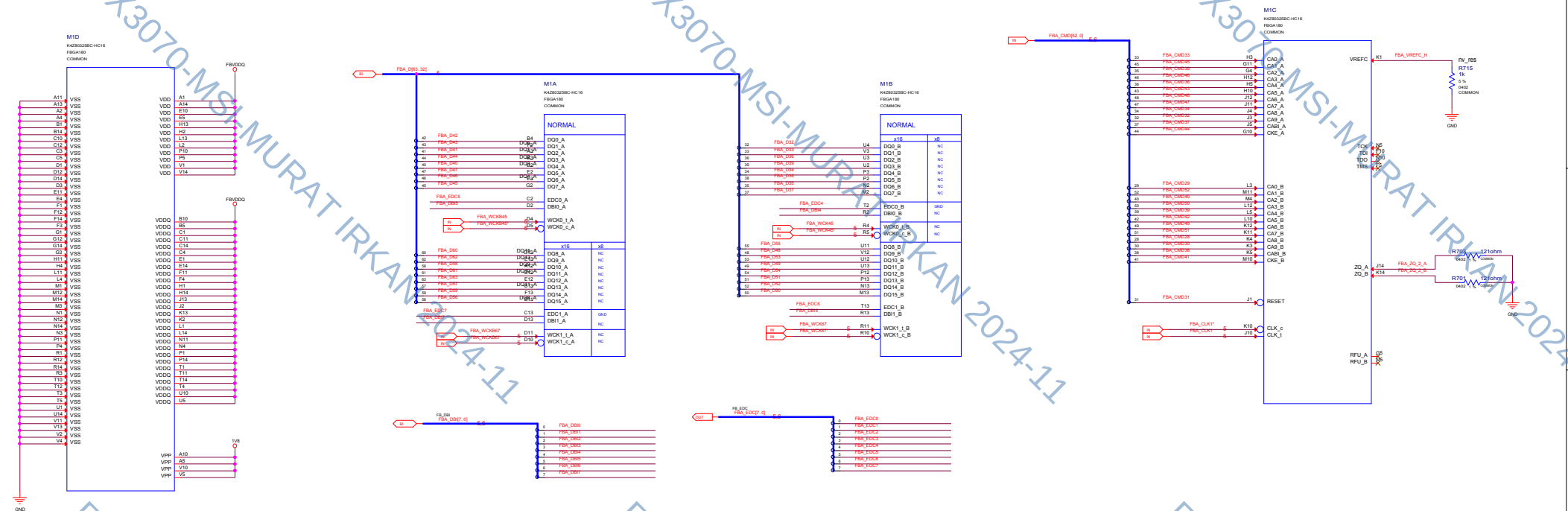
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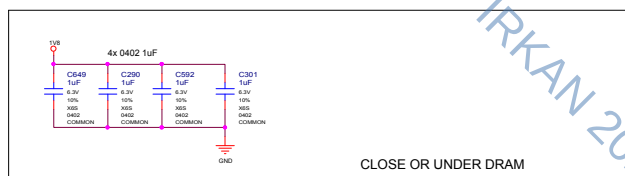
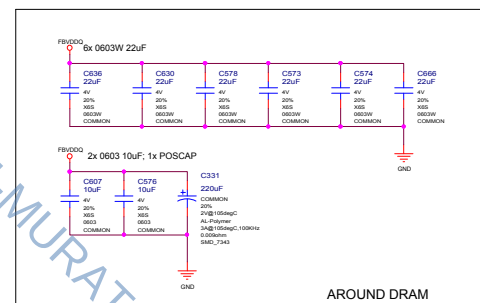
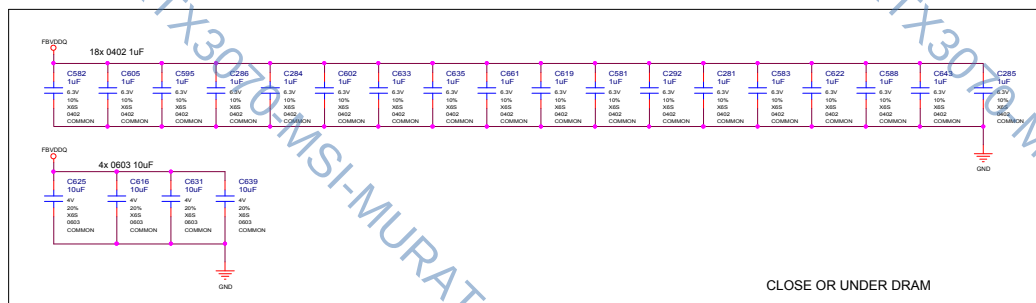
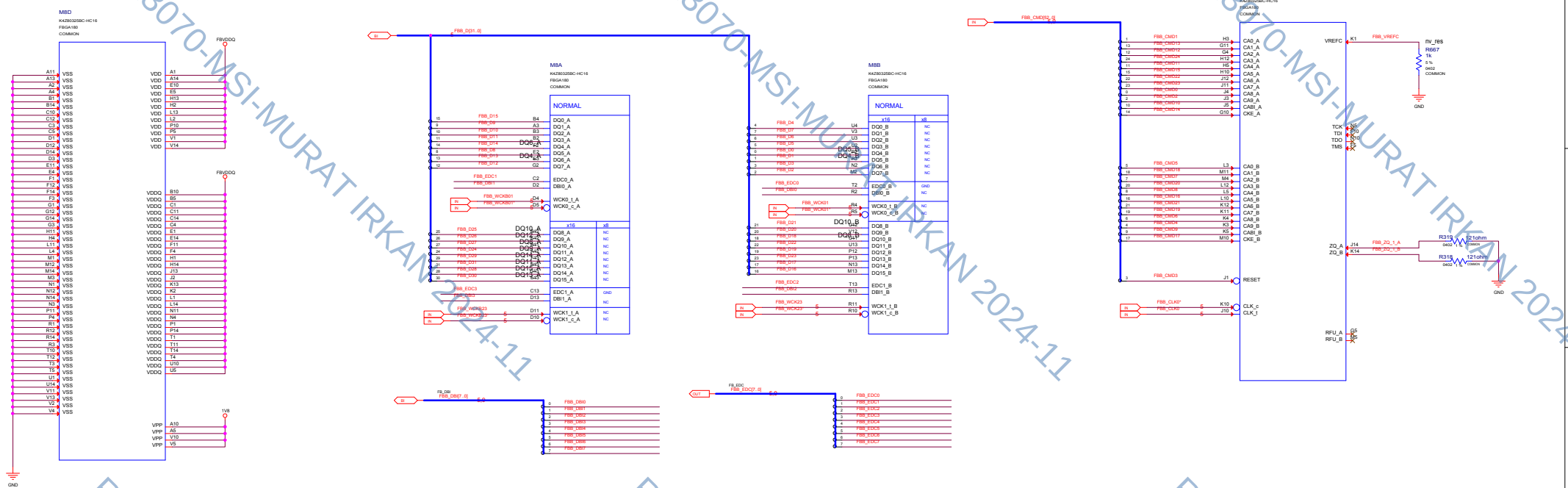
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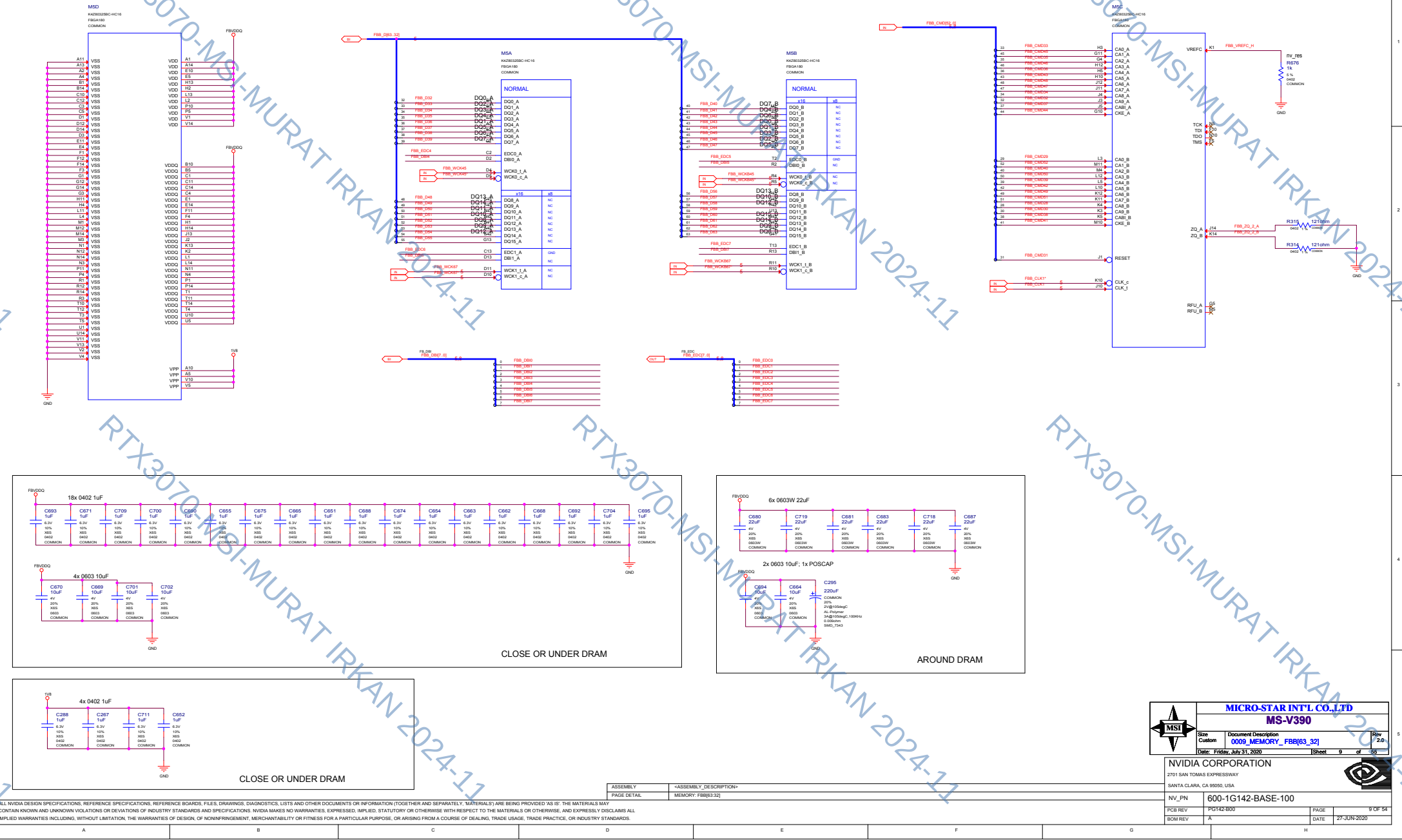


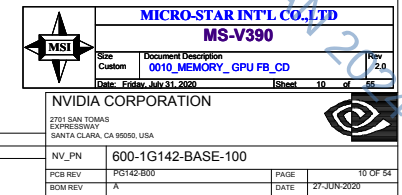
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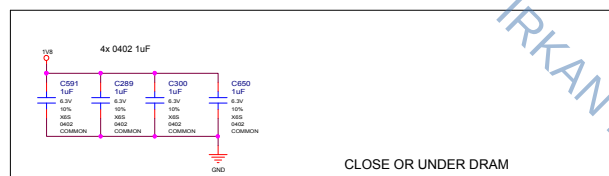
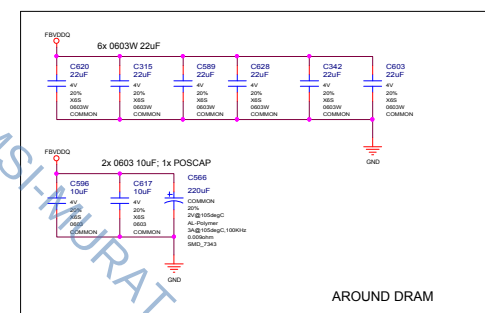
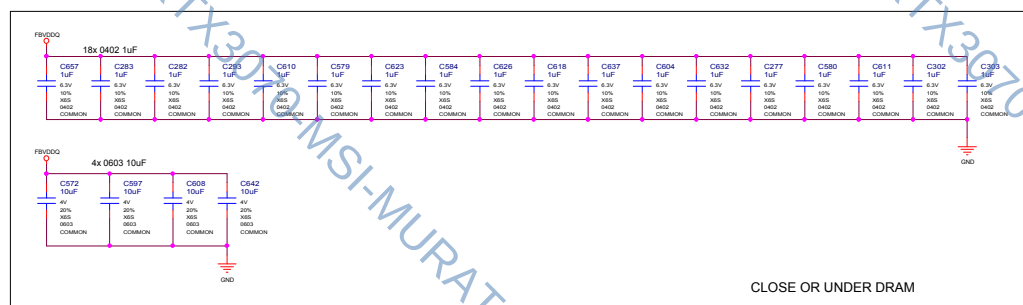
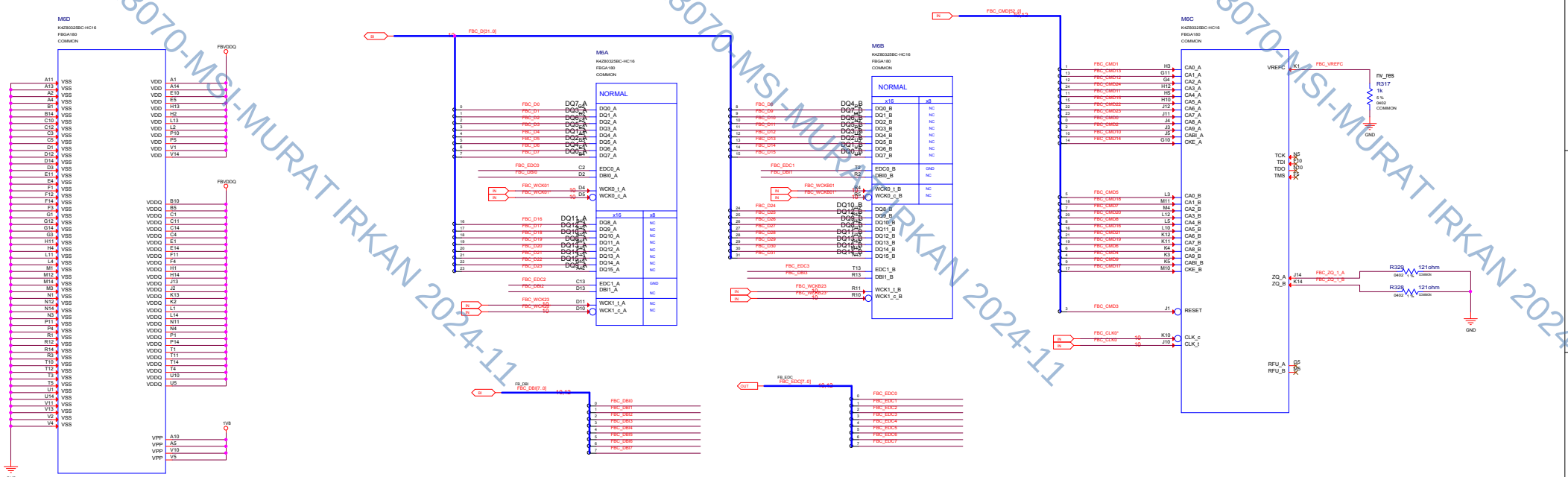
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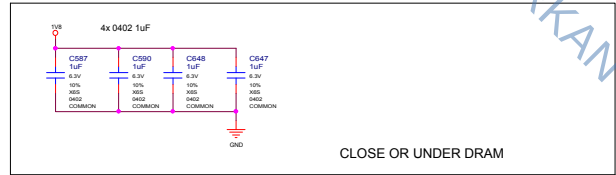
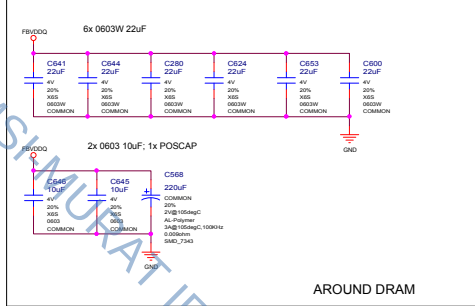
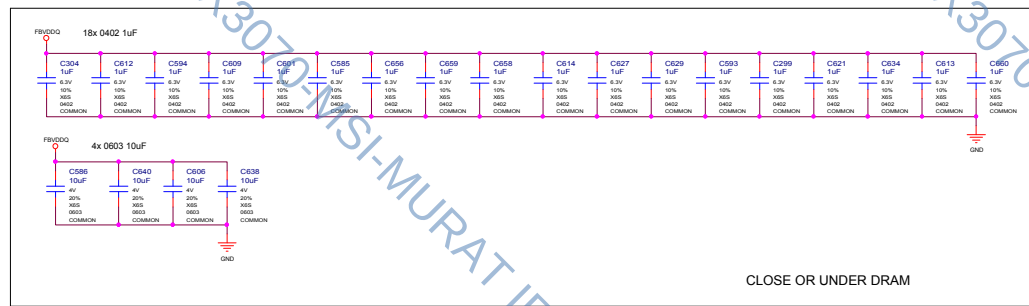
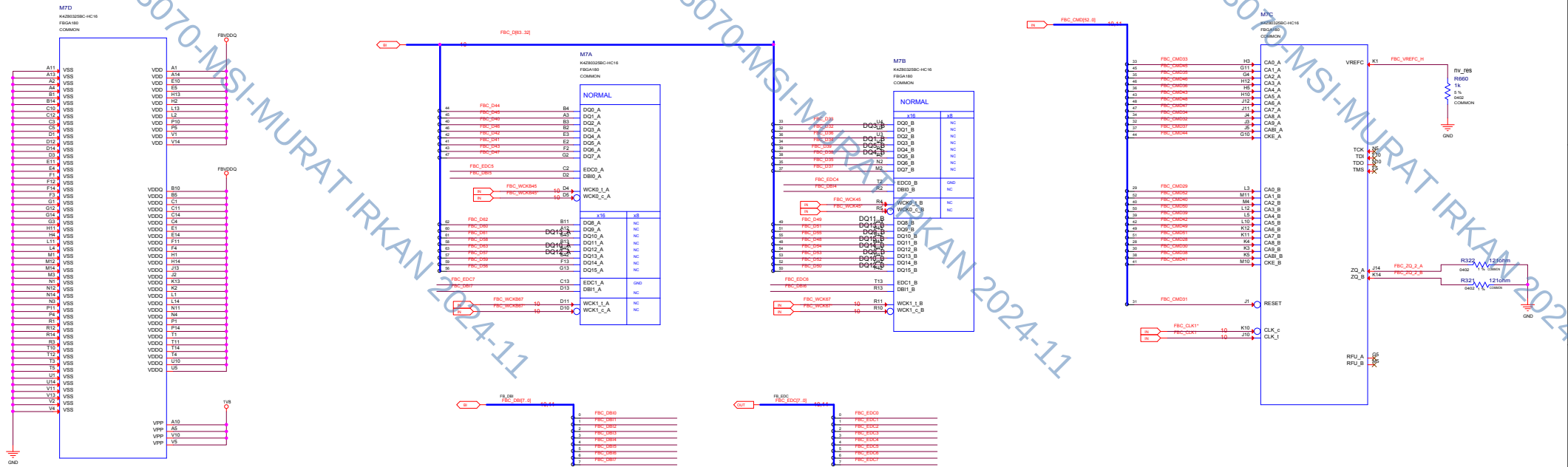




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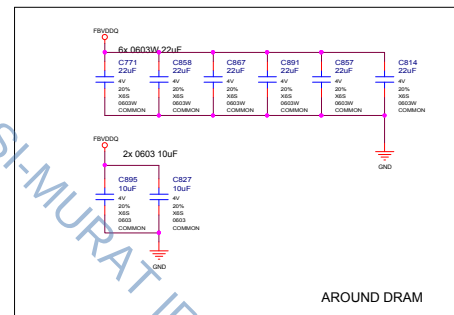
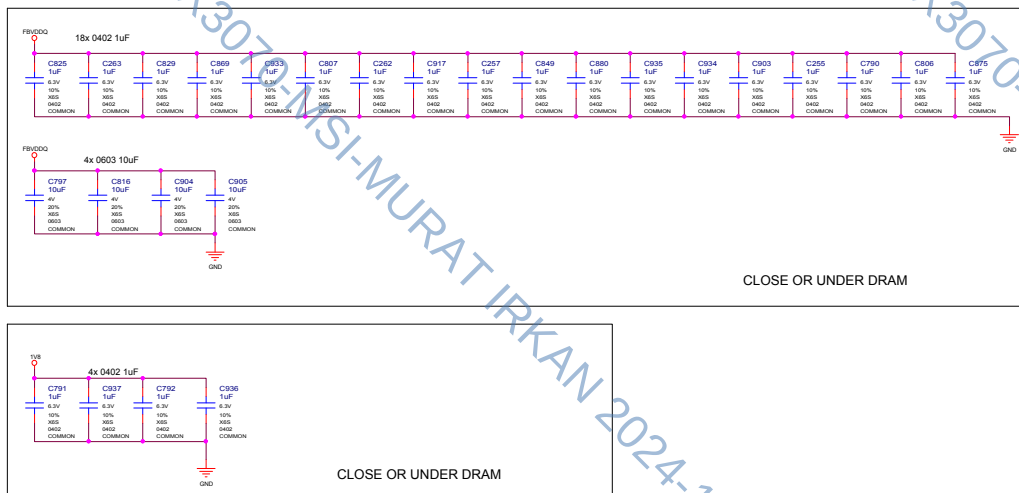
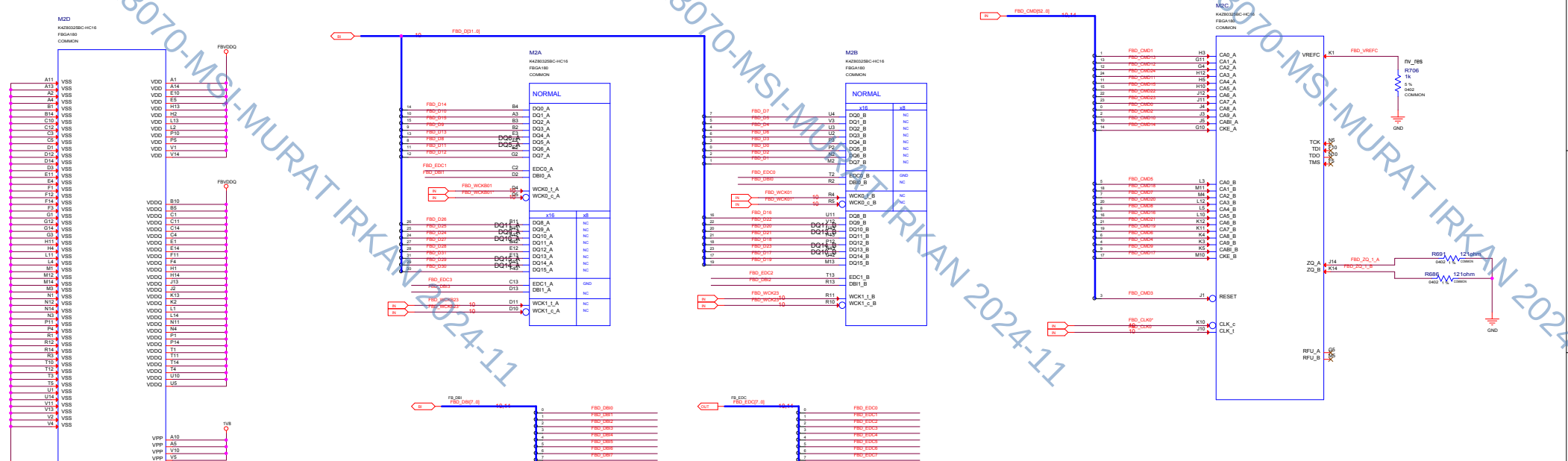


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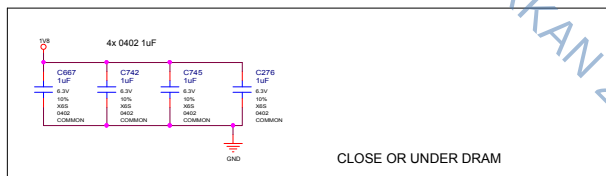
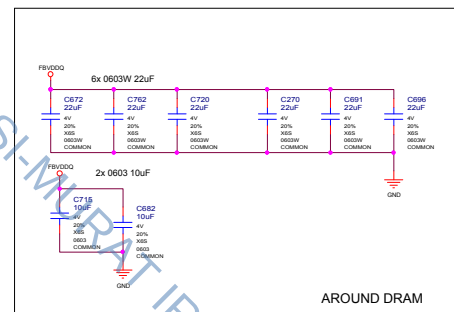
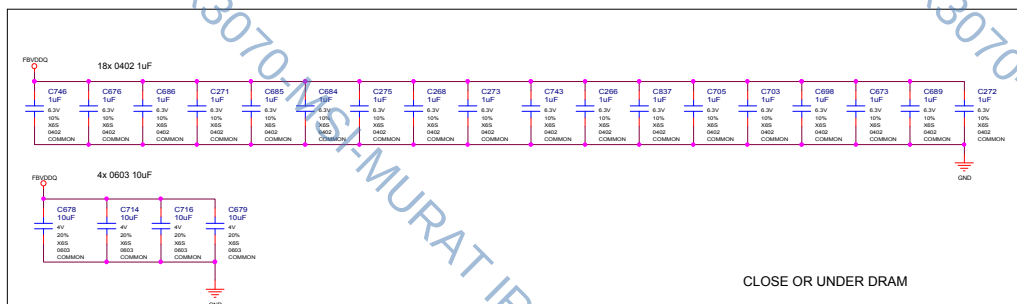
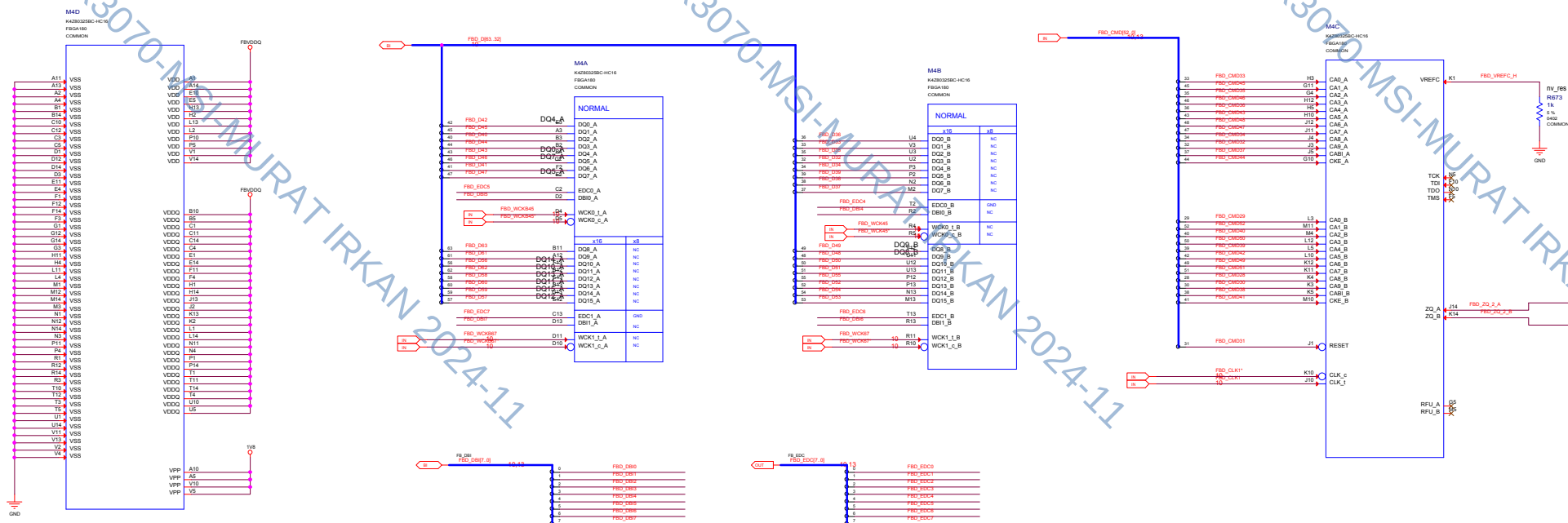
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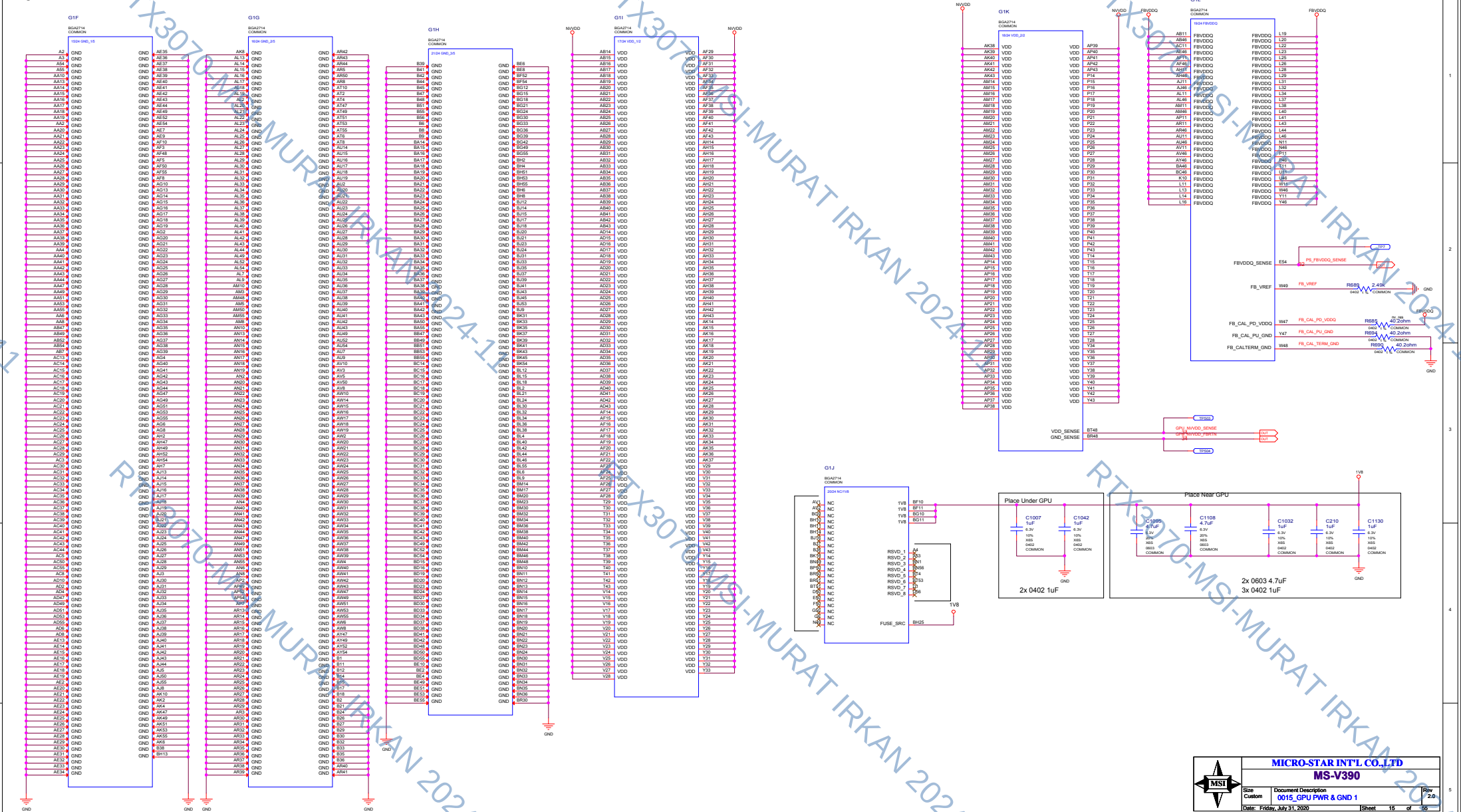
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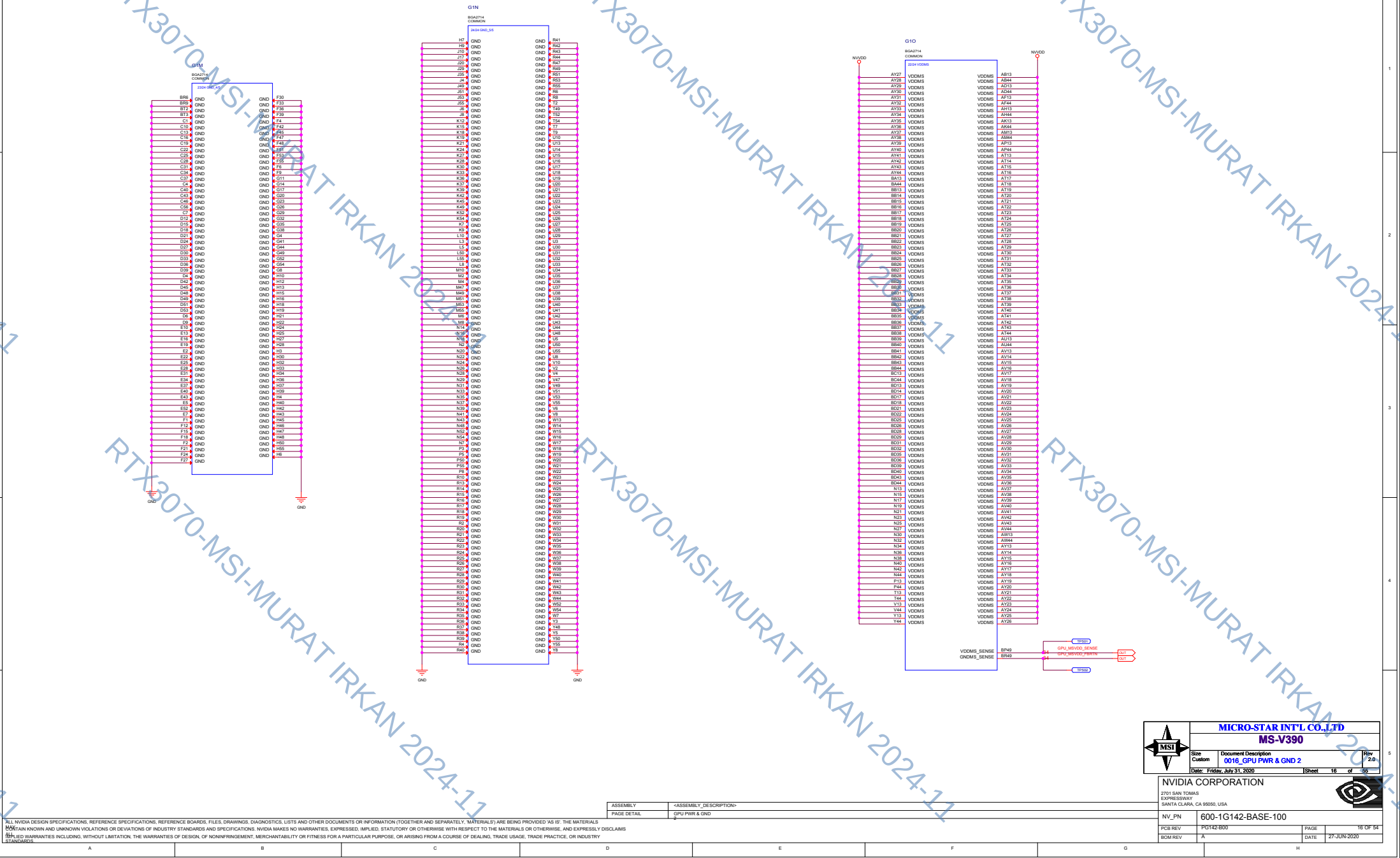
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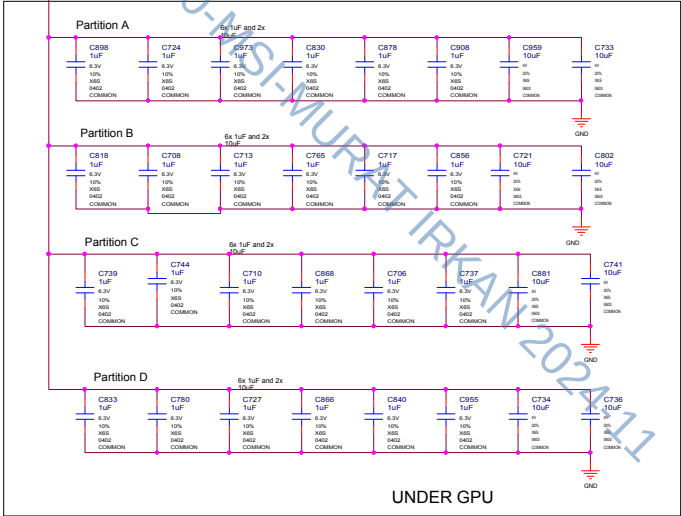
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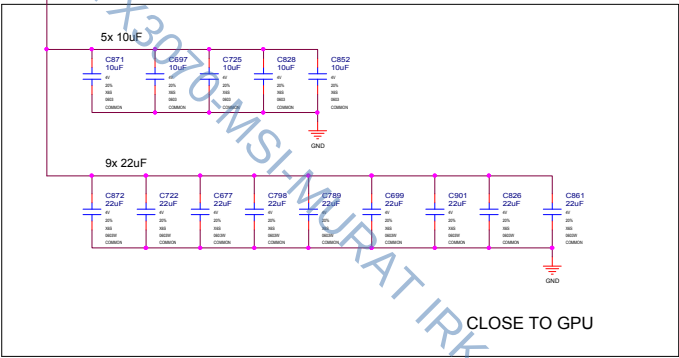
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UNDER GPU

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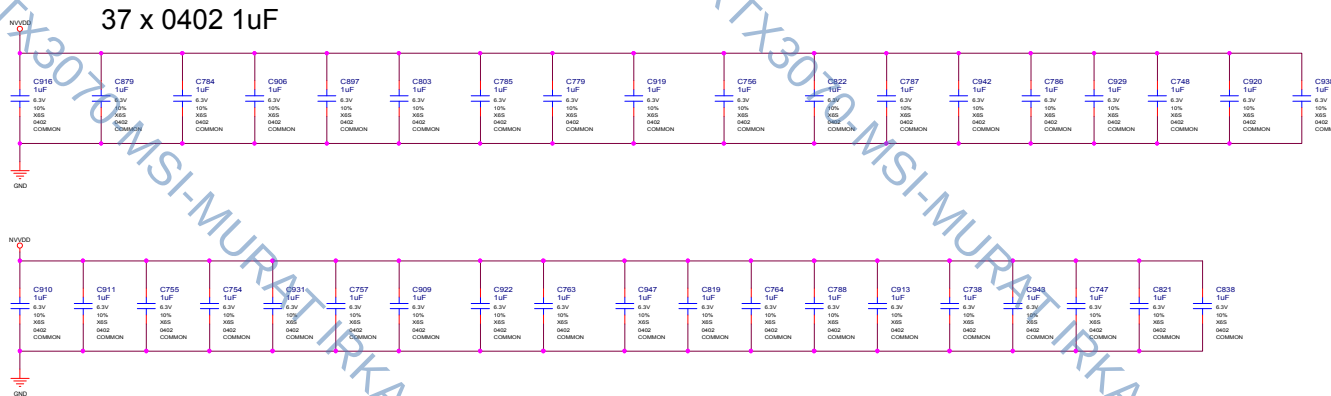
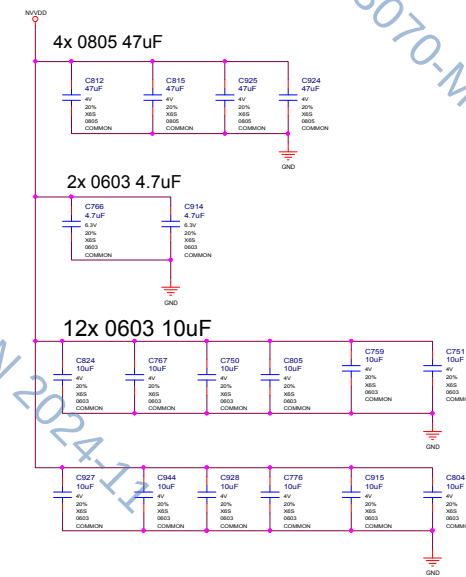
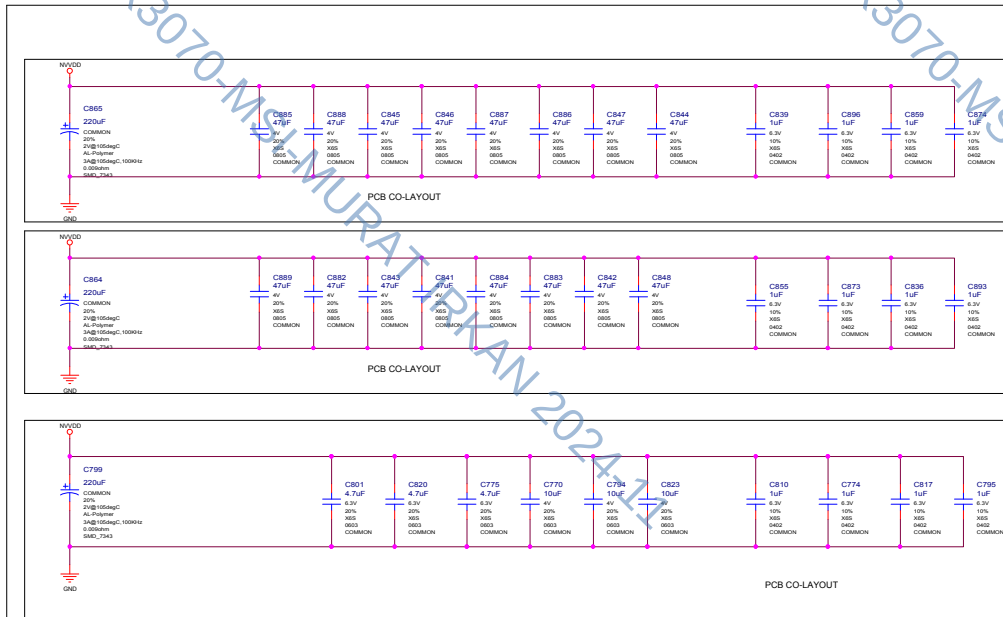


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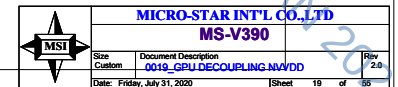
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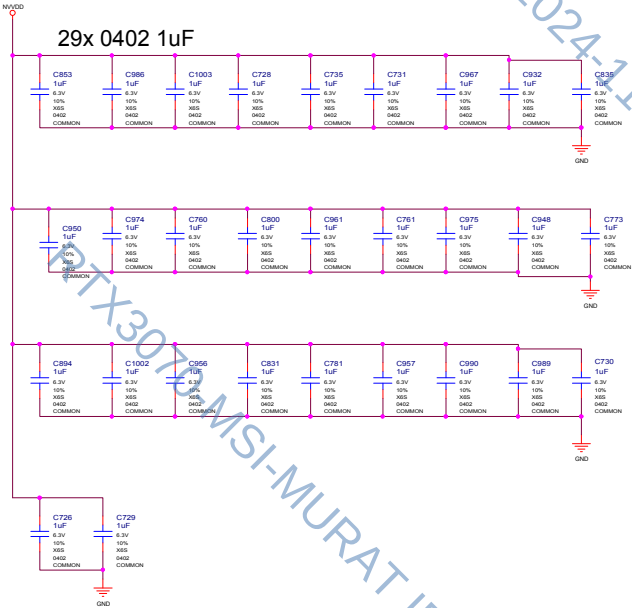
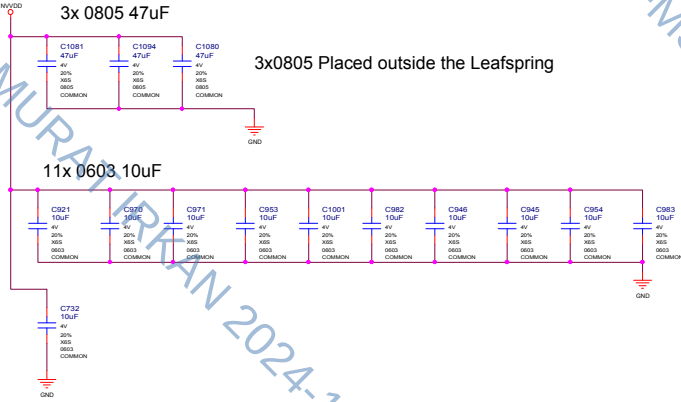
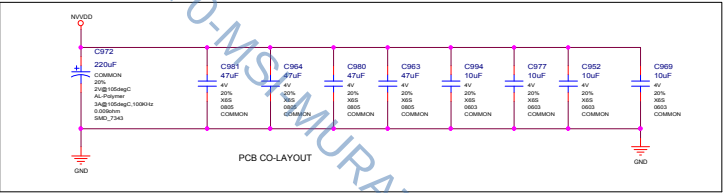


NV_PN	600-1G142-BASE-100
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[illegible]

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UNDER GPU



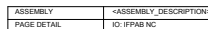
MICRO-STAR INT'L CO., LTD			
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Date:	Friday, July 31, 2020	Sheet	20 of 55

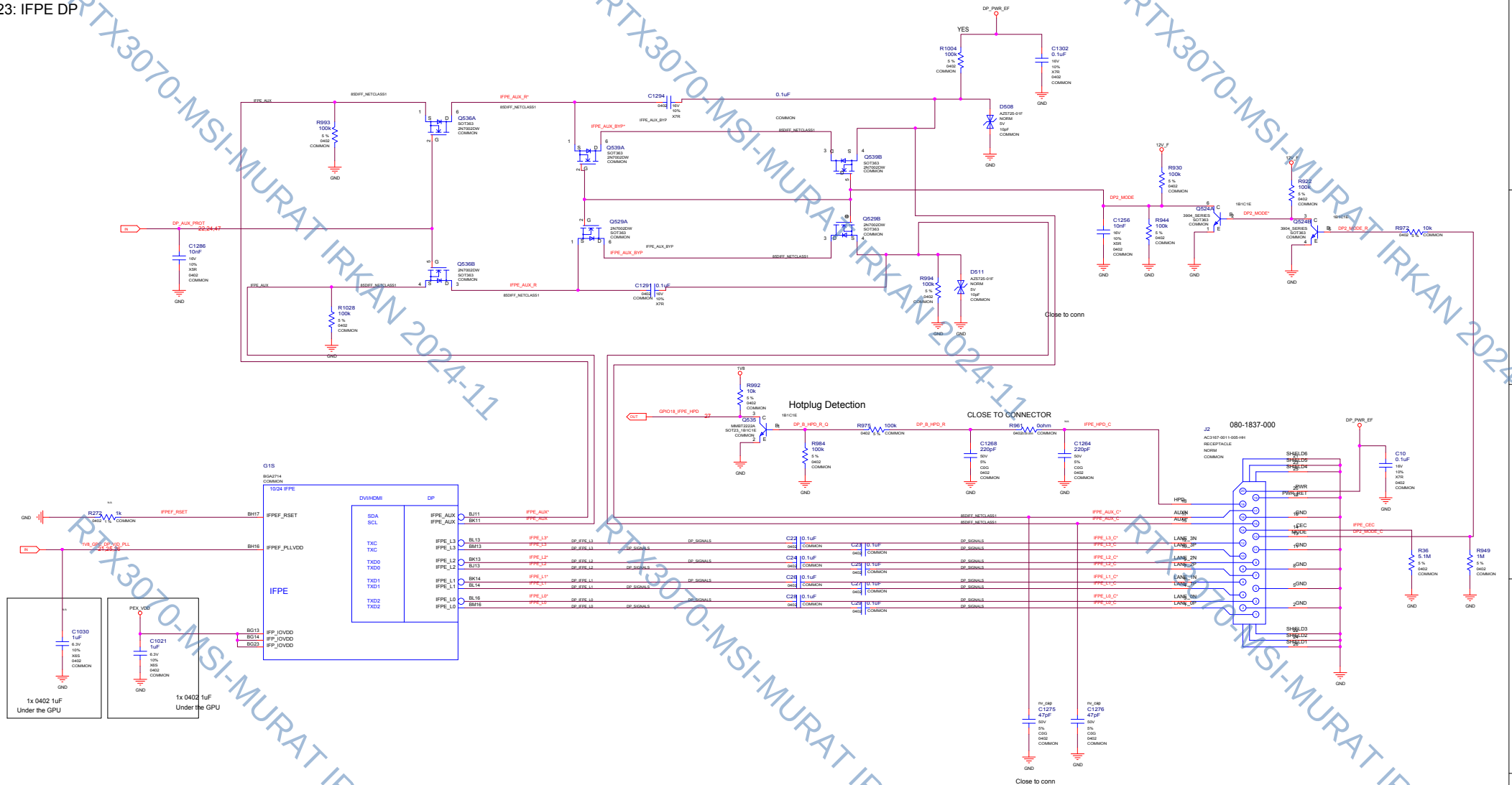


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	Cus: F616W_IPFE_21_2020 Sheet 23 of 26			
NVIDIA CORPORATION 2701 SAN TOMAS EXPRESSWAY SANTA CLARA, CA 95050, USA				
				
NV_PN		600-1G142-BASE-100		

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DATE	A	DATE	2/7-20/2020	

STRAP2	STRAP1	STRAP0	RAMCFG[4:0]	
L	L	L	00000	RAMCFG SAMSUNG 8Gb 7GHz
L	L	H	00001	RAMCFG MICRON 8Gb 7GHz
L	H	L	00010	RAMCFG HYNIX 8Gb 7GHz
H	H	L	00110	RAMCFG SAMSUNG 16Gb 8GHz
TBD				

H=High :Tied to 1.8V
M=Middle:Tied to 0.9V
L=Low :Tied to 0V

ROM_SO	ROM_SI	ROM_SCLK	SMARTFAN[2:0]FS_OVERT	1:ENABLE 0:DISABLE	DEFAULT
H	H	H	0111	FS_OVERT ENABLE	
H	H	M	0000	FS_OVERT DISABLE	

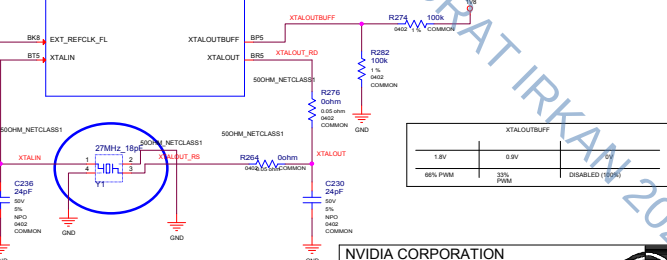
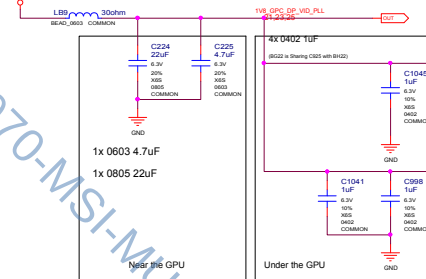
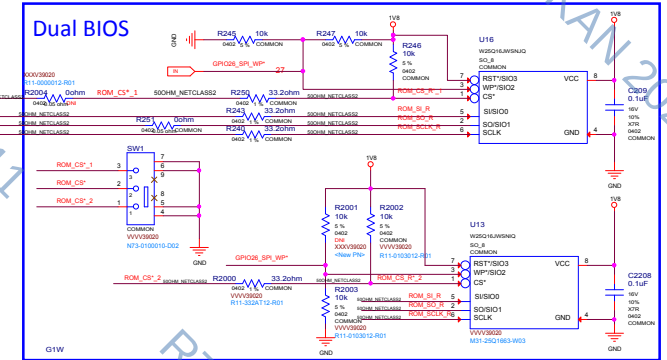
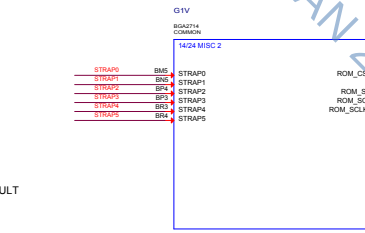
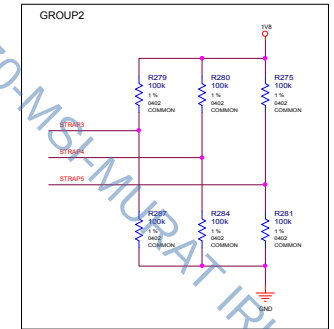
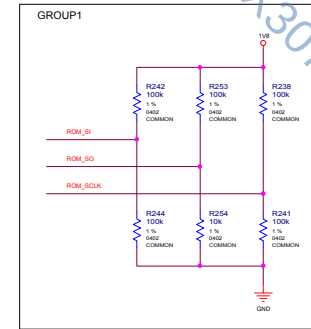
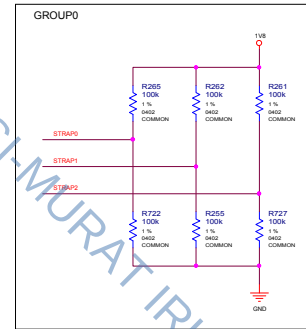
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M	H	H	1	1	1	1
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M	L	H	1	1	0	1
M	L	L	1	1	0	0
L	H	M	1	0	1	1
L	M	L	1	0	0	1
L	L	M	1	0	0	0
H	H	H	0	1	1	1
H	H	L	0	1	1	0
H	L	H	0	1	0	1
H	L	L	0	1	0	0
L	H	H	0	0	1	1
L	H	L	0	0	1	0
L	L	H	0	0	1	0
L	L	L	0	0	0	1 DEFAULT

H=High :Tied to 1.8V
M=Middle:Tied to 0.9V
L=Low :Tied to 0V

1:SMB_ALT_ADDR ENABLE
0:SMB_ALT_ADDR DISABLE
1:DEVID_SEL REBRAND
0:DEVID_SEL ORIGINAL

1:PCIE_CFG LOW POWER
0:PCIE_CFG HIGH POWER
1:VGA_DEVICE ENABLE
0:VGA_DEVICE DISABLE

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Size	Document Description	Rev	
Custom	0026_MISC1_ROMx2L_XTAL_STRAPS	2.0	
Date: Friday, July 31, 2020	Sheet 26 of 55		
ASSEMBLY	ASSEMBLY DESCRIPTION		
PAGE DETAIL	MISC1: ROM, XTAL, STRAPS		

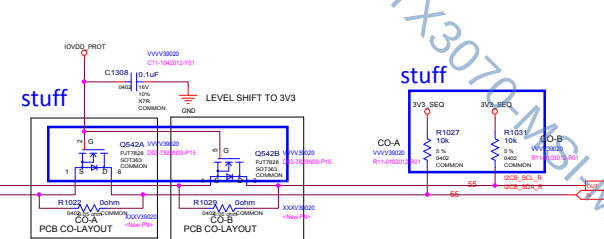


XTALOUT	XTALOUT	XTALOUT
1.8V	0.9V	DISABLED
80% PWM	20% PWM	DISABLED

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NV_PN	600-1G142-BASE-100	PAGE	26 OF 54
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BIOM REV	A		



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STEREO
remove STEREO

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	Friday, July 31, 2009		Sheet 27 of 54	
NVIDIA CORPORATION 2701 SAN TOMAS EXPRESSWAY SANTA CLARA, CA 95050, USA				
				
NV_PLN		600-1G142-BASE-100		
PB0 REV	PS142-800	PAGE	27 OF 54	
FORM REV	A	DATE	27-JUN-2000	



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MS-V390

Size	Custom	Document Description	0026_MISC3_RGB	Rev	2.0
Date:	Friday, July 31, 2020			Sheet	28 of 66

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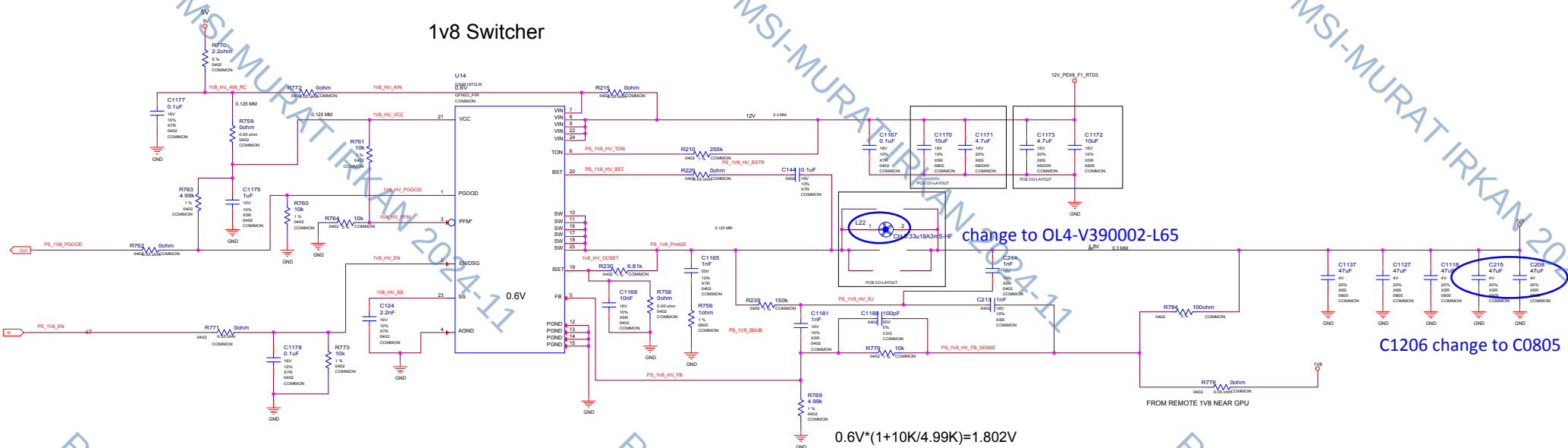
2701 SAN TOMAS EXPRESSWAY
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NV_PN	600-1G142-BASE-100		
PCB REV	PG142-B00	PAGE	28 OF 64
BOM REV	A	DATE	27-JUN-2020

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1v8 Switcher



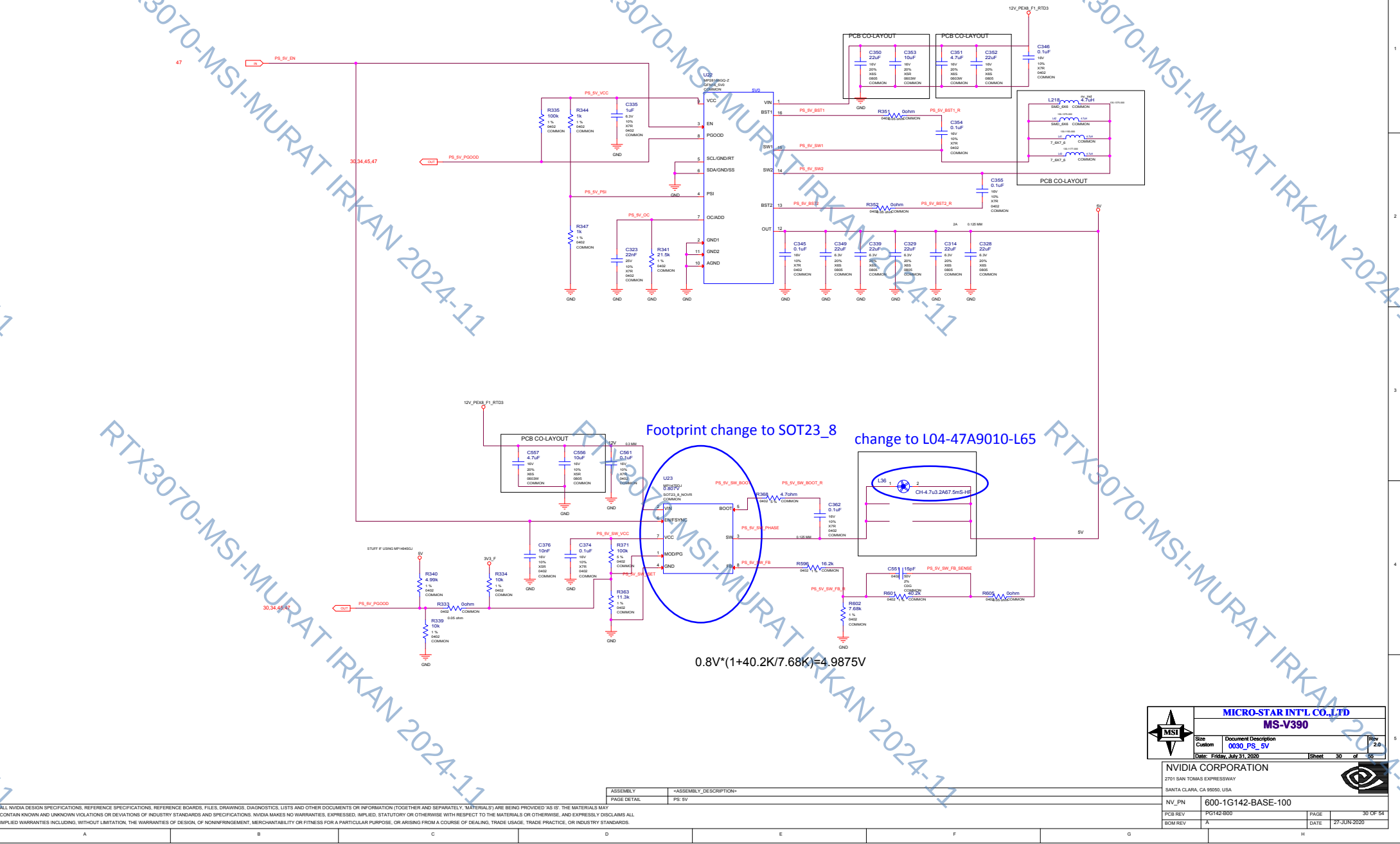
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change to OL4-V390002-L65

C1206 change to C0805

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Date:	0029_PS_V1v6			No.	
Friday, July 31, 2020	Sheet	29	of	30	
NVIDIA CORPORATION					
2701 SAN TOMAS EXPRESSWAY					
SANTA CLARA, CA 95050, USA					
					
NV, NVL	600-1G142-BASE-100				
TCB REV	P0142-000	PAGE	28 OF 54		
1008 REV	A	DATE	27-JUN-2020		

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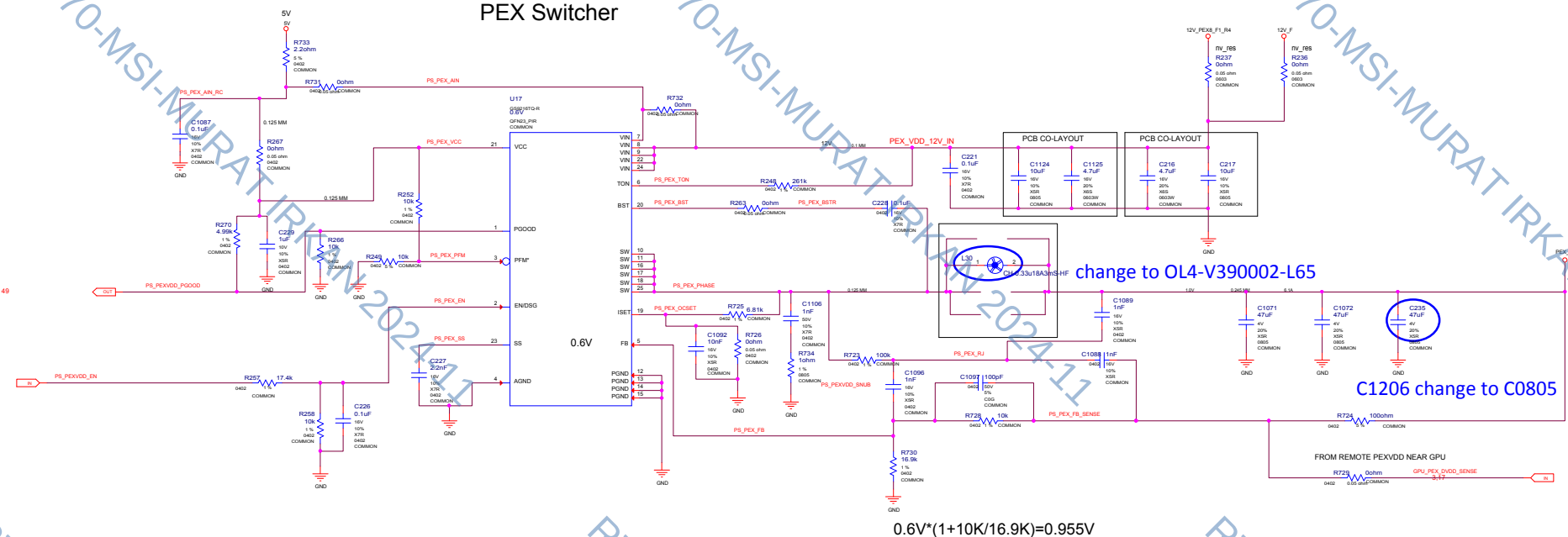
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Date: Friday, July 31, 2020		Sheet	30 of 36

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ICOM REV	A	DATE	27-JUN-2020

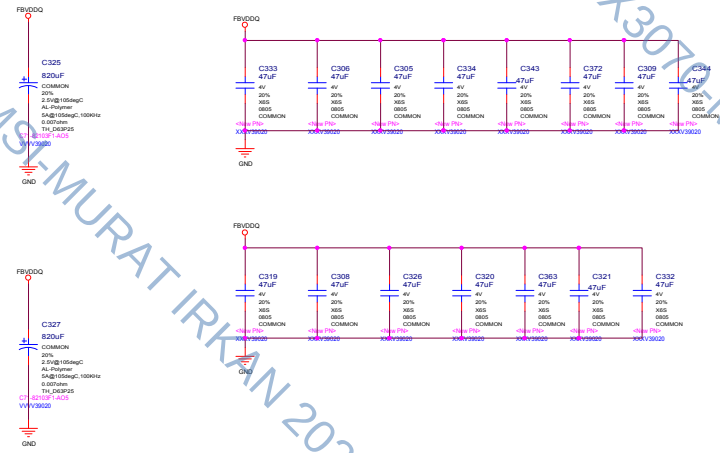
PEX Switcher



MICRO-STAR INT'L CO., LTD			
MS-V390			
Size	Document Description	Rev	2.0
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Date:	Friday, July 31, 2020	Sheet	31 of 36

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debug, not stuff.

delete TP13~15

RT1 change to R51-0104T12-M09
delete RT2

add PH11

add PH11

change to I32-816100C-005
footprint : QFN40_TSMQ142

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Date: Friday, July 31, 2020		Sheet	34 of 56

add PH11

I2C_Add = 0x4A

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NCV_PN 600-1G142-BASE-100

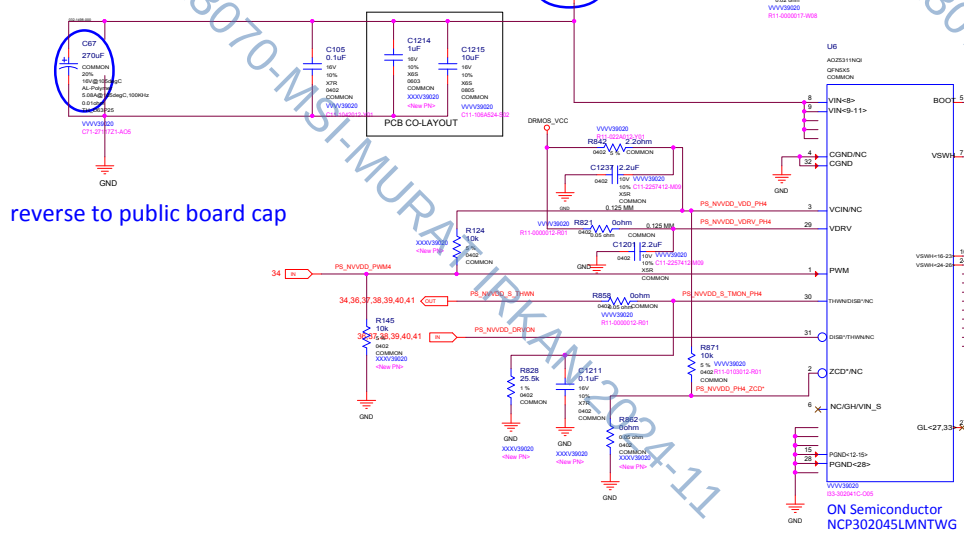
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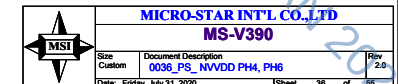
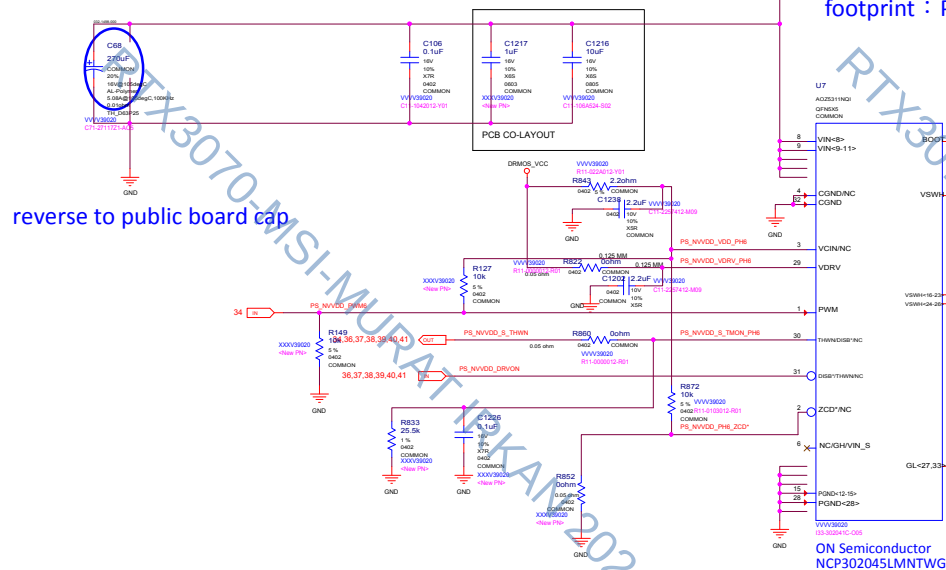
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Date: Friday, July 31, 2020		Sheet	35 of 54
NVIDIA CORPORATION			
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SANTA CLARA, CA 95050, USA			
NV_PN	600-1G142-BASE-100		
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NVVDD PH4(PWM4)



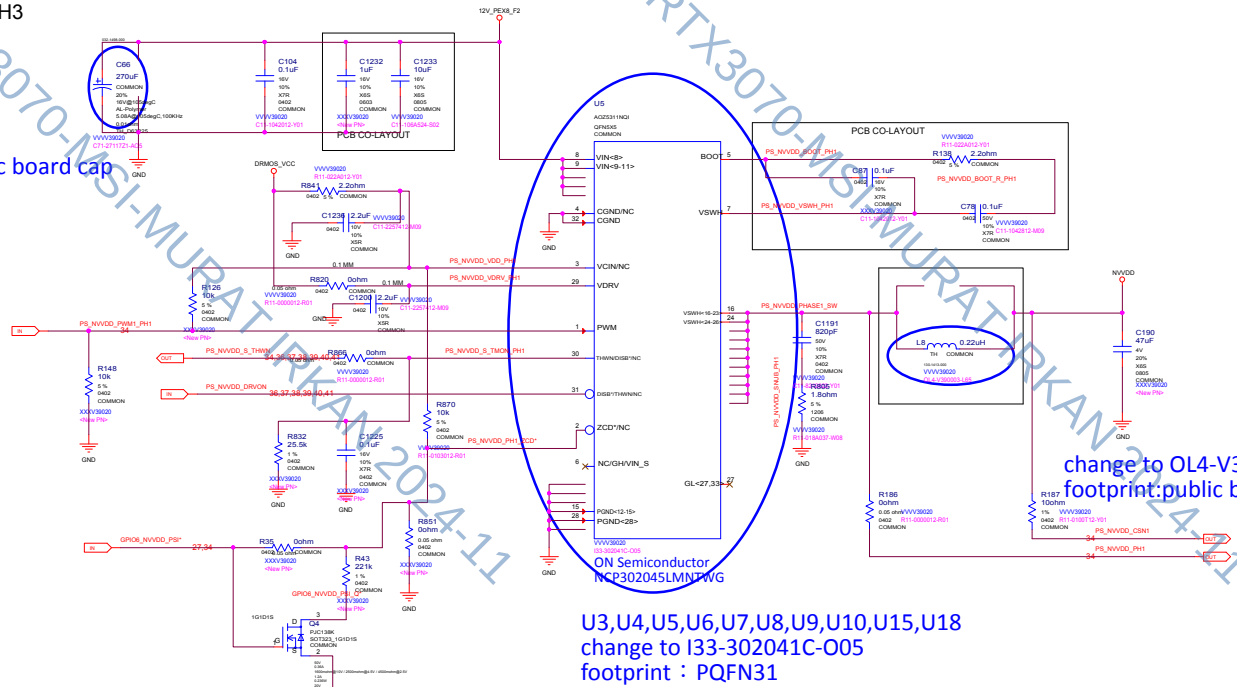
NVVDD PH6(PWM6)



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NV_PN	600-1G142-BASE-100
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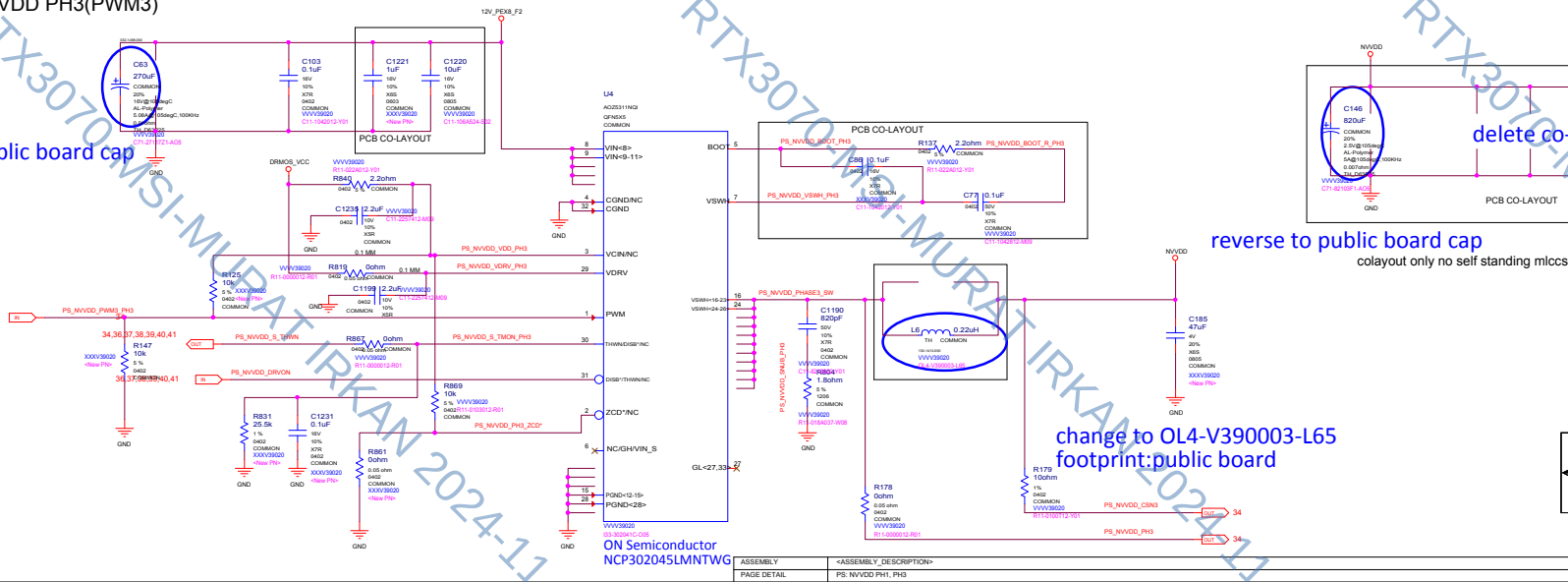
NVVDD PH1(PWM1)

reverse to public board cap



NVVDD PH3(PWM3)

reverse to public board cap

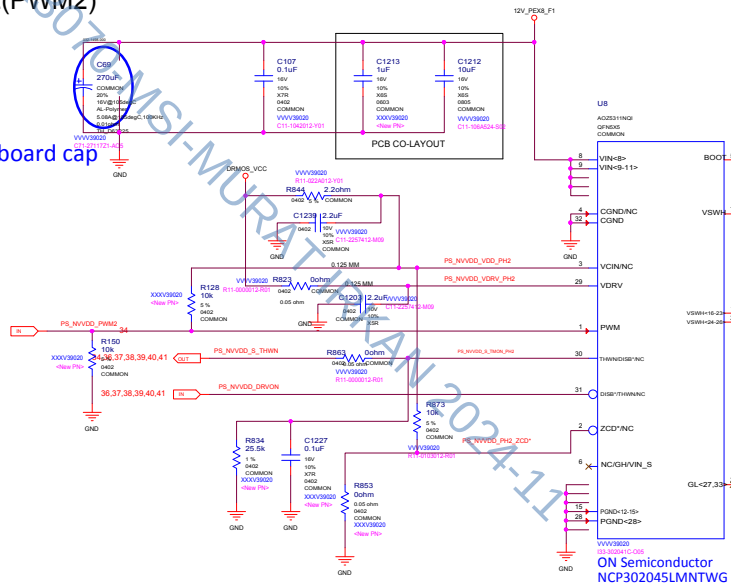


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MS-V390			
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Date: Friday, July 31, 2020		Sheet	37 of 55
NVIDIA CORPORATION			
2701 SAN TOMAS EXPRESSWAY			
SANTA CLARA, CA 95050, USA			
NV_PN		600-1G142-BASE-100	
PCB REV	PG142-B00	PAGE	37 OF 54
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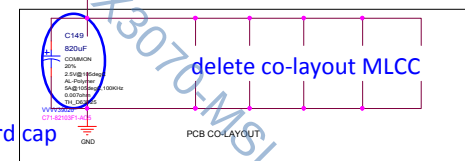
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NVVDD PH2(PWM2)

reverse to public board cap

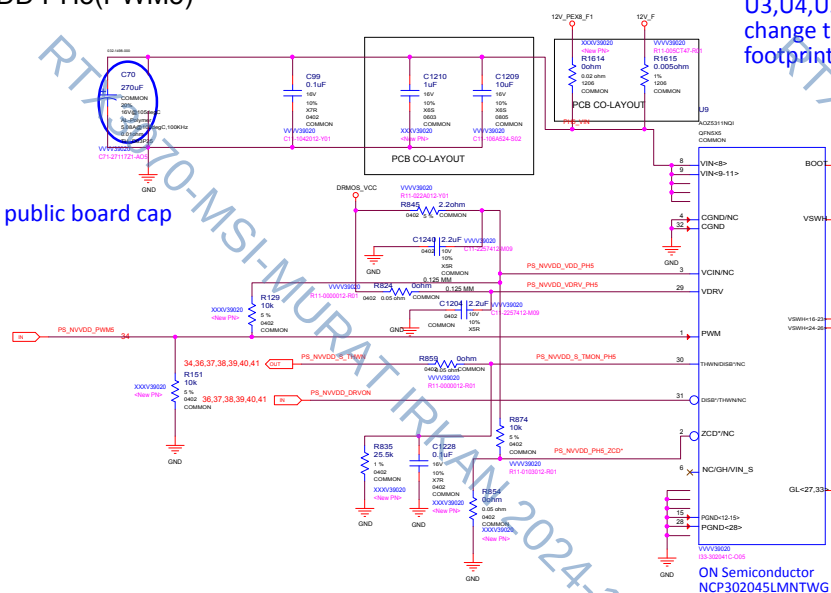


reverse to public board cap

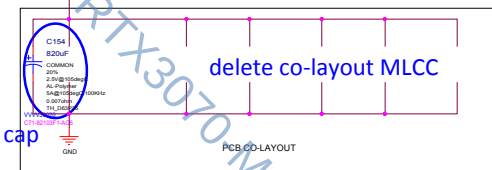
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footprint:public board

NVVDD PH5(PWM5)

reverse to public board cap

U3,U4,U5,U6,U7,U8,U9,U10,U15,U18
change to I33-302041C-O05
footprint : PQFN31

reverse to public board cap

change to OL4-V390003-L65
footprint:public board

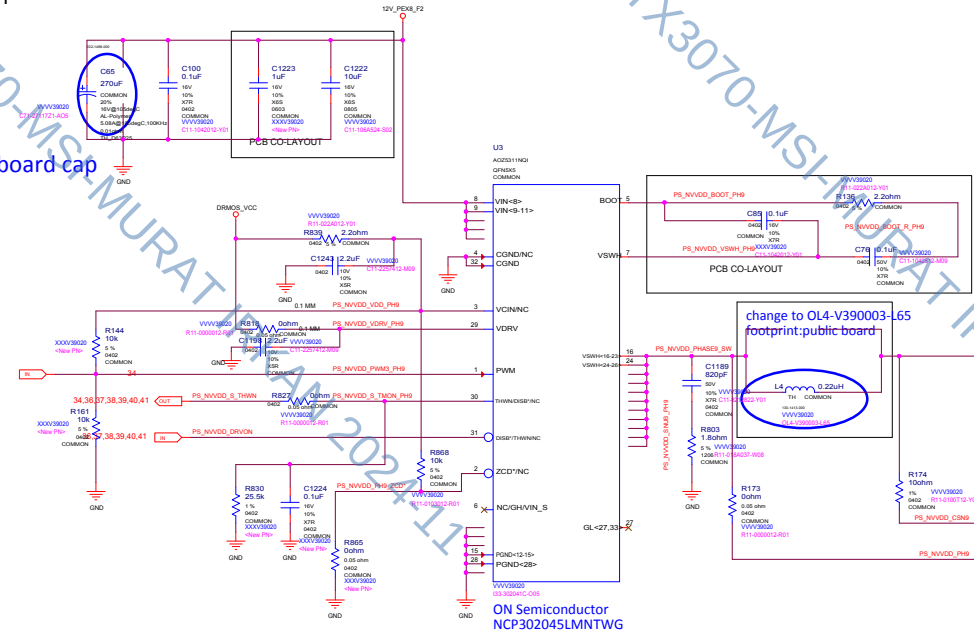
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PAGE DETAIL	PS_NVVDD PH2, PH5

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MS-V390			
Size	Document Description	Rev	
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Date:	Friday, July 31, 2020	Sheet	38 of 36
NVIDIA CORPORATION			
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NV_PN	600-1G142-BASE-100		
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NVVDD PH9(PWM3)

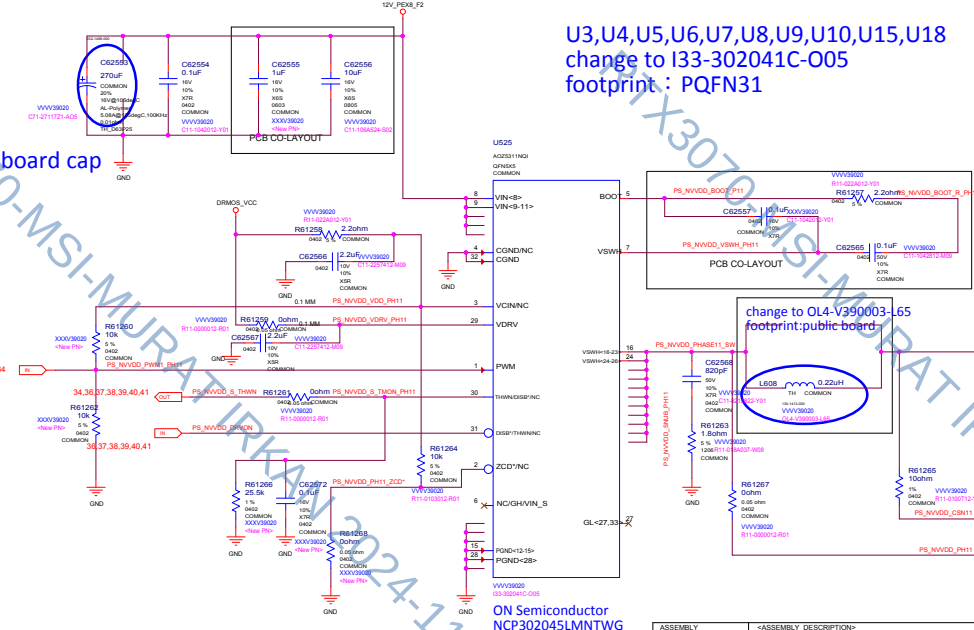
reverse to public board cap



ON Semiconductor
NCP30204SLMNTWG

NVVDD PH11(PWM1)

reverse to public board cap



ON Semiconductor
NCP30204SLMNTWG

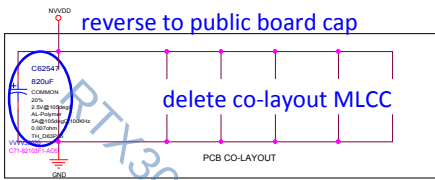
reverse to public board cap

delete co-layout MLCC



reverse to public board cap

delete co-layout MLCC





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Size	Custom	Document Description	0030 PS_NVVDD PH9, PH11	Rev	2.0
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NVVDD PH7(PWM7)

reverse to public board cap

reverse to public board cap

delete co-layout MLCC

change to OL4-V390003-L65
footprint:public board

ON Semiconductor
NCP302045LMNTWG

U3,U4,U5,U6,U7,U8,U9,U10,U15,U18
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footprint: PQFN31

MICRO-STAR INT'L CO.,LTD				
MS-V390				
Size	Custom	Document Description	0040_PS_NVVDD PH7	Rev 2.0
Date:	Friday, July 31, 2020	Sheet	40	of 55

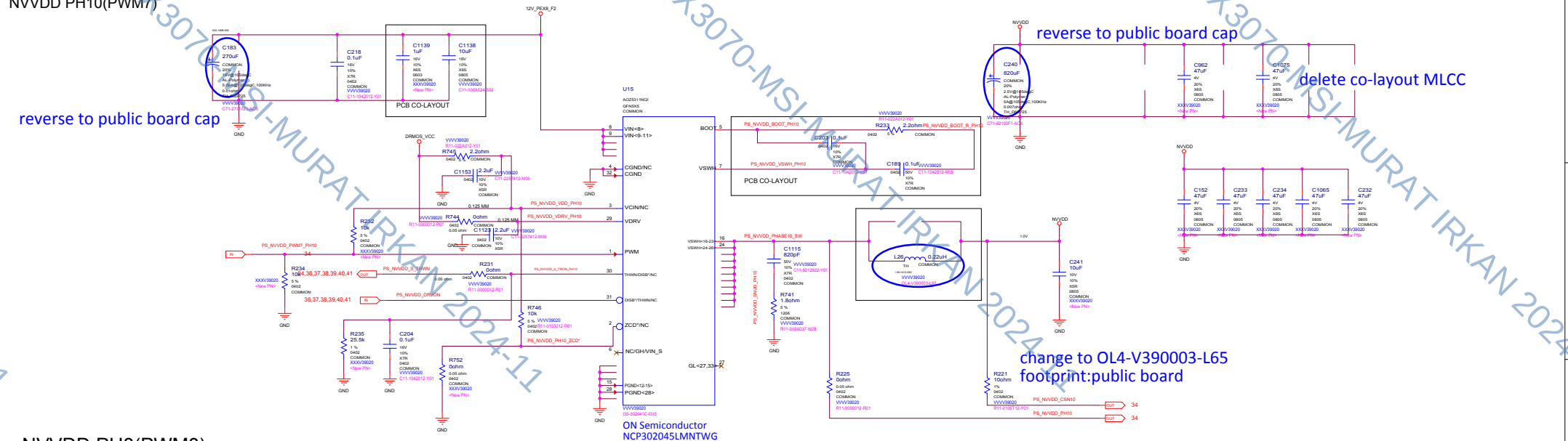
NVidia CORPORATION
2701 SAN TOMAS EXPRESSWAY
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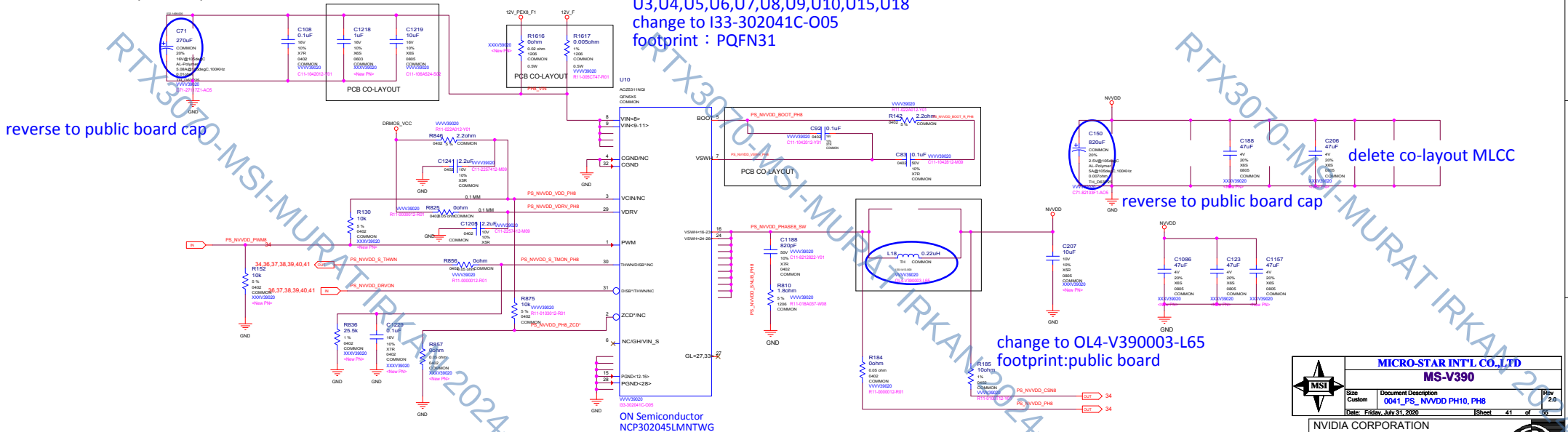
NV_PN	600-1G142-BASE-100
PCB REV	PG142-B00
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ASSEMBLY	<ASSEMBLY_DESCRIPTION>
PAGE DETAIL	PS_NVVDD PH7

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NVVDD PH8(PWM8)

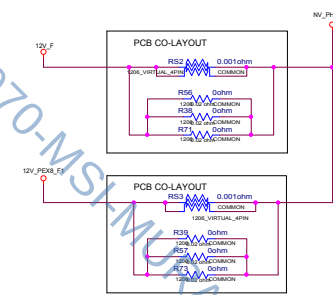
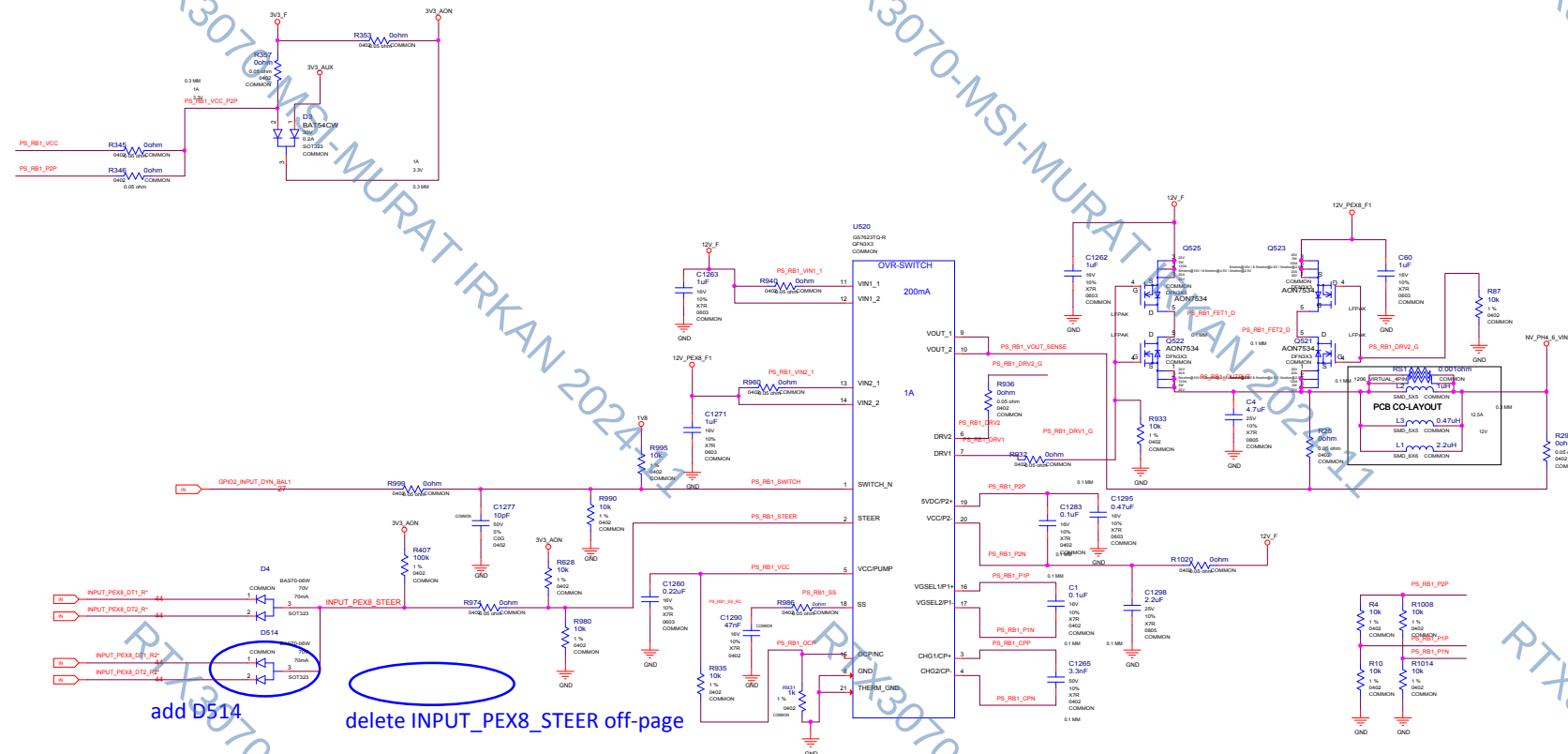


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		0041_PS_NVDD PH10, PH8				2
	Date:	Friday, 2001	Sheet	41	of 50	

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		2/7/2000



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	MS-V390		
Size Custom	Document Description 0042_PS_OVRS_INPUT POWER BALAN		Rev 2.0
Date: Friday, July 31, 2020		Sheet 42 of 65	

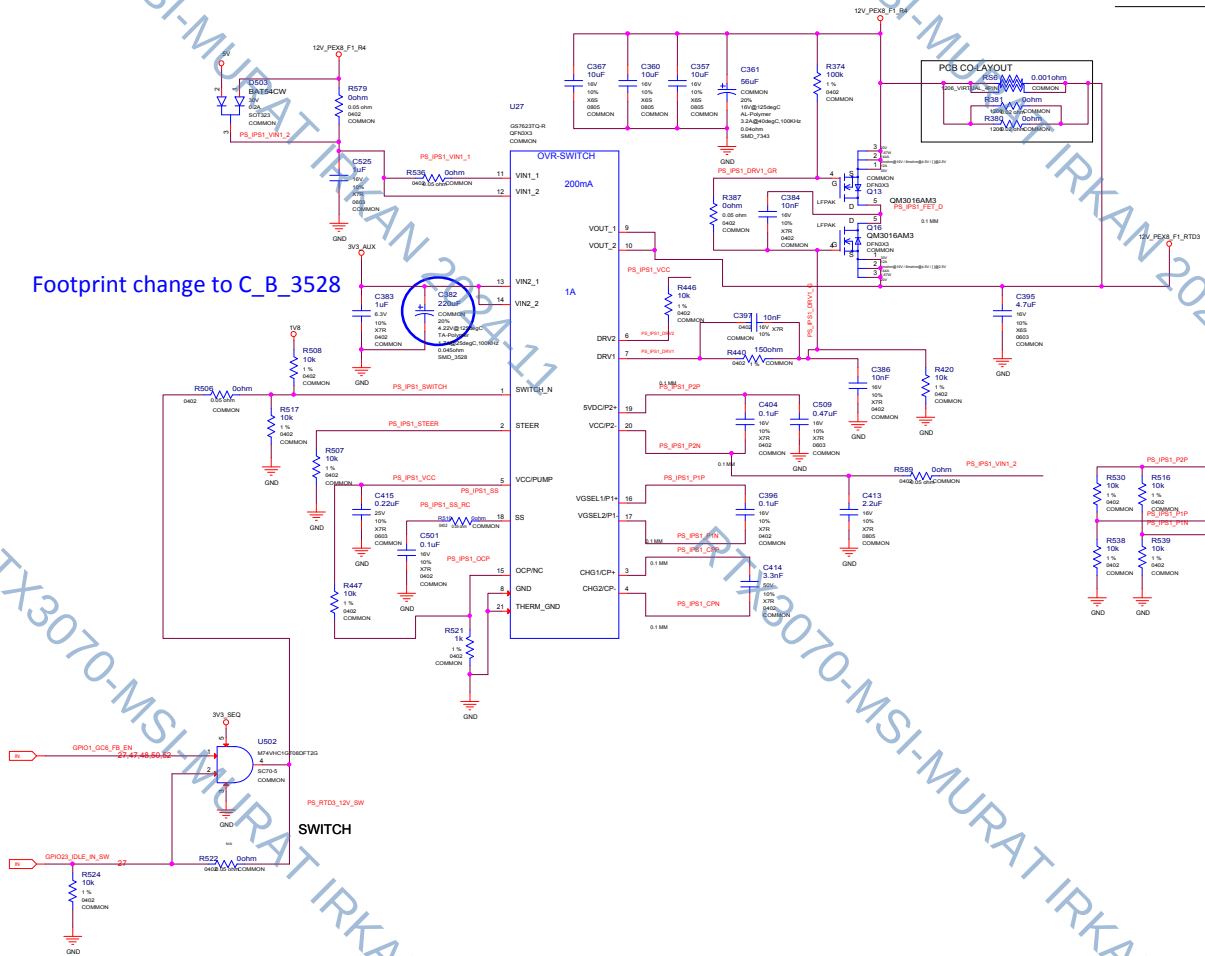
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NV_PN	600-1G142-BASE-100		
PCB REV	PG142-B00	PAGE	42 OF 54
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AND GATE LOGIC				
GPIO1	GPIO29	SWITCH	VOUT	VOUT
0	0	0	12V	VOUT = VIN1
0	1	0	12V	VOUT = VIN2
1	0	0	12V	
1	1	1	3V3A	



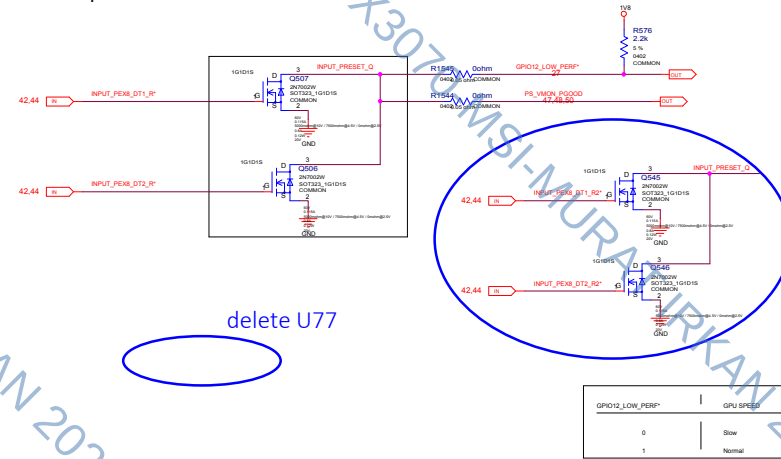
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Date: Friday, July 31, 2020		Sheet	43	of	56

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delete 12 pin 12V EXT-POWER Current steering

PEX Input - Under Power Boot Control



delete U77

PEX8 INPUT 2/B - 2x4 PCIE CON 150W

Fuse

PCB CO-LAYOUT

F7 FUSE-2A,5610269-HF

12.5A

delete R2007

Fuse

PCB CO-LAYOUT

F9 FUSE-2A,5610269-HF

12.5A

delete R2010

PEX8 INPUT 1/A - 2x4 PCIE CON 150W

delete R2010

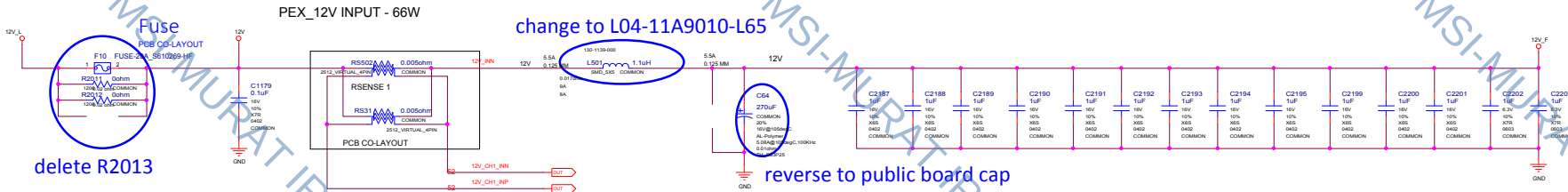
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add RS24

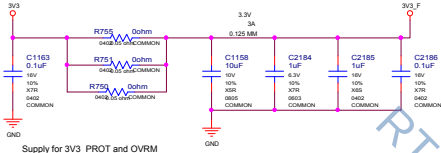
change to L04-33B7320-L65

one 12pin EXT power conn. change to two reverse 8pin conn.

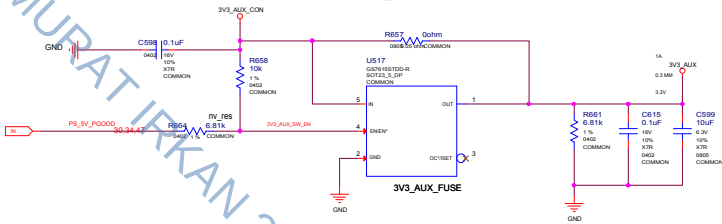
MICRO-STAR INT'L CO., LTD			
MS-V390			
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Custom	0044_PS_INPUTS, FILTERING and	1	2
Date: Friday, July 31, 2020		Sheet	44 of 55
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PEX 3V3 INPUT - 10W



3V3_AUX



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Custom	0045_P8_PEX_12V, PEX_3V3 AND A	2.0	
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ASSEMBLY	<ASSEMBLY_DESCRIPTION>
PAGE DETAIL	PEX_12V, PEX_3V3 AND 3V3_AUX

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POWER CONNECTOR HOT-UNPLUG DETECT

delete POWER CONNECTOR HOT-UNPLUG DETECT

12V CURRENT STEERING (UNDER POWER BOOT):

GUIDES CURRENT FROM PEX EDGE TO PEX 8 PIN INPUT AREA

delete 12V CURRENT STEERING



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Date:	Friday, July 31, 2020		Sheet	46	of 56

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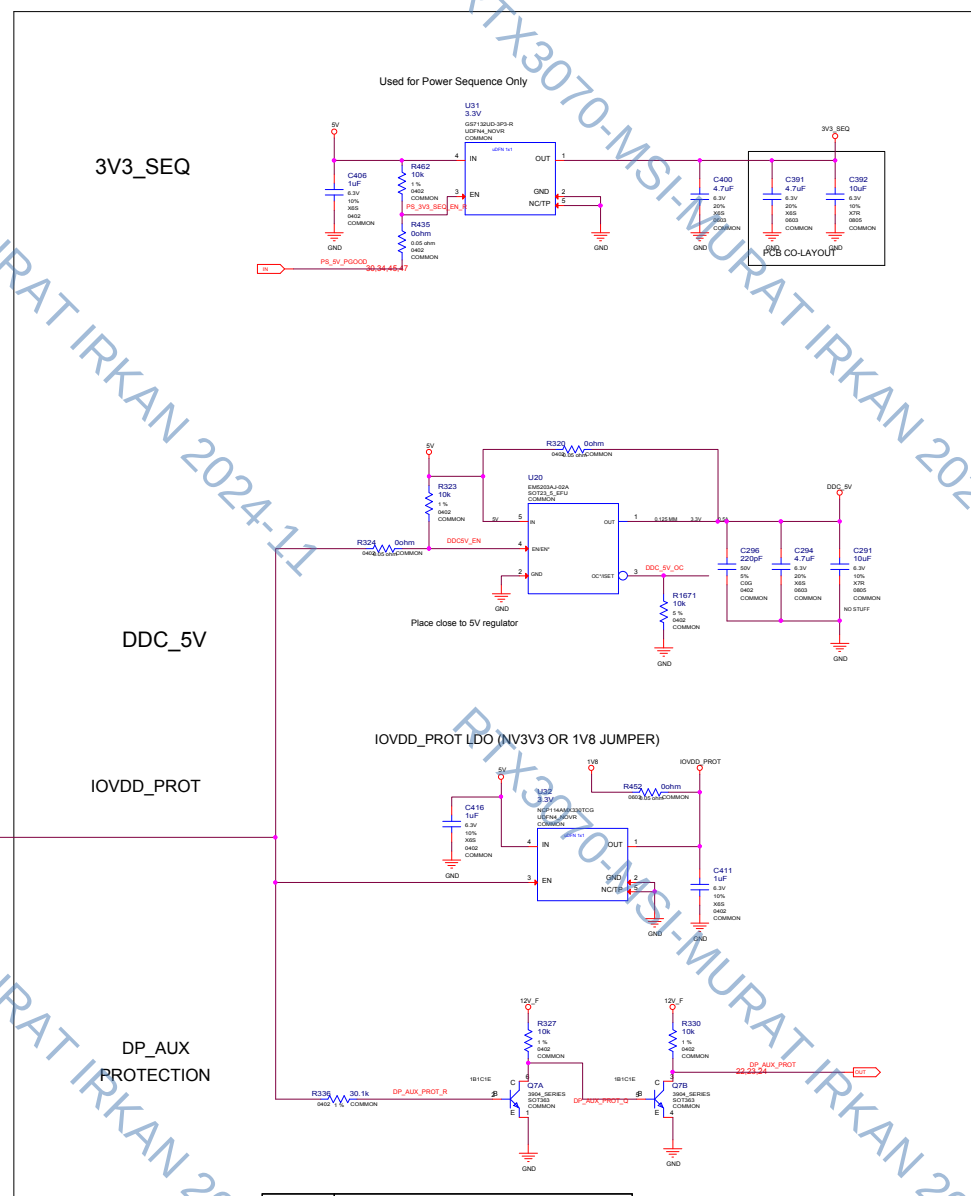
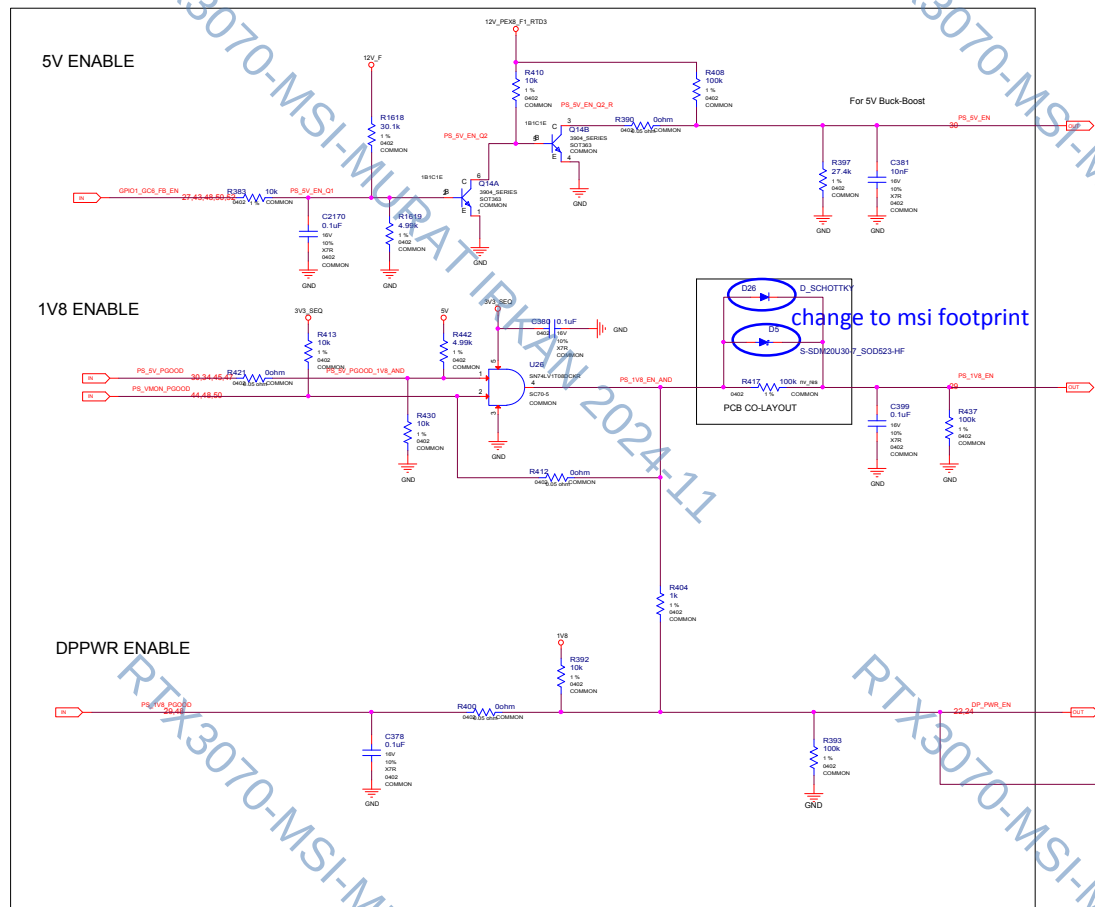


NV_PN

600-1G142-BASE-100

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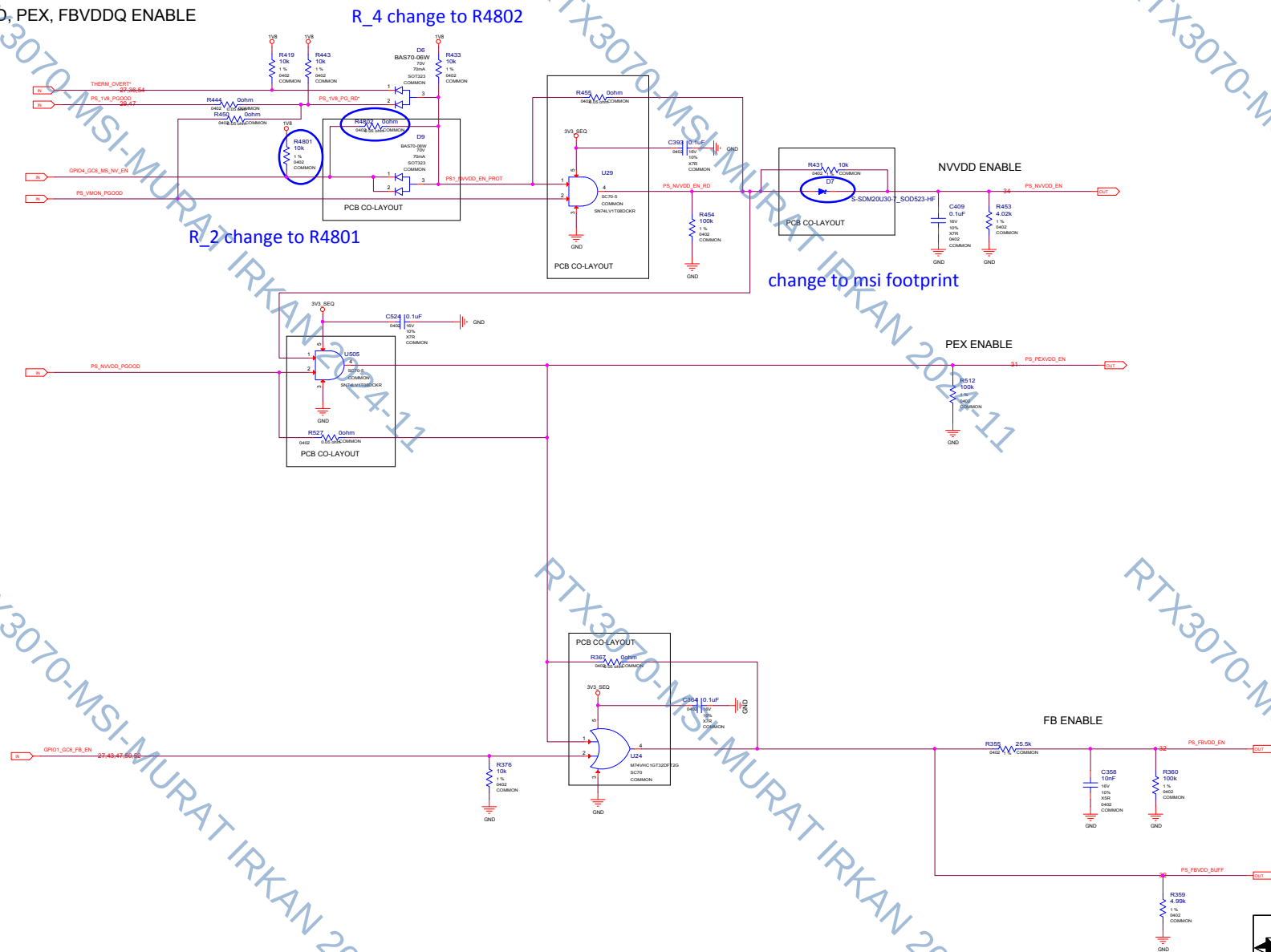
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Date: Friday, July 31, 2020		Sheet 47 of 66	

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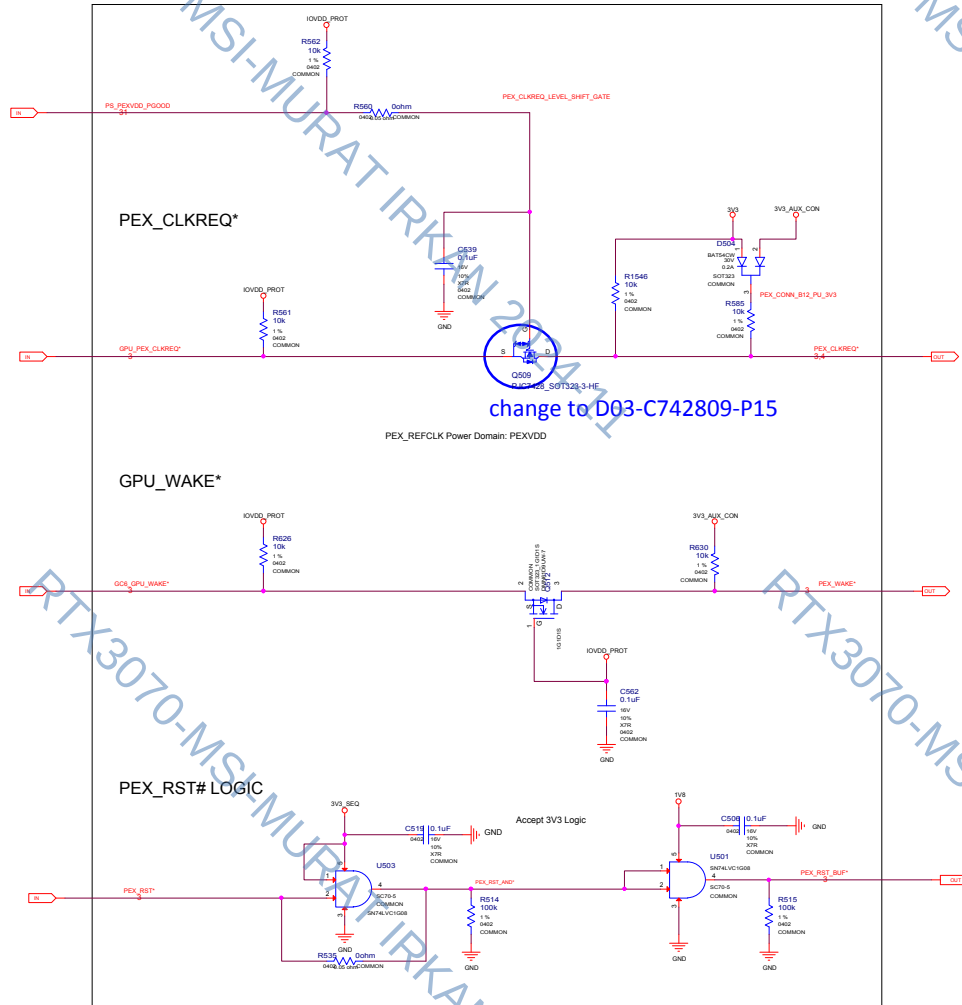


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	Size	Document Description	Rev
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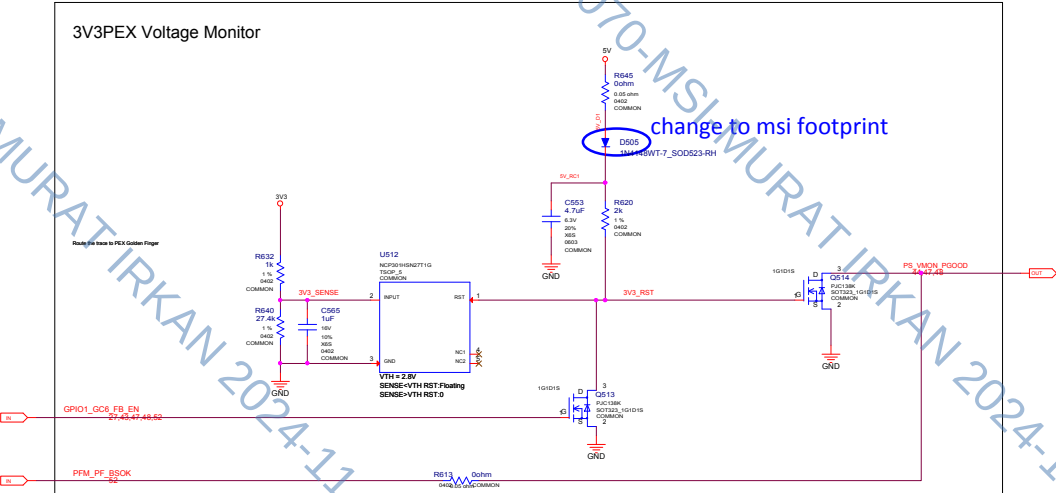
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Custom	0049_SEQ_MISC	2.0	
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BOM Configuration

OPTIONS	PEX3V3_SENSE	PEX12V_SENSE	OTHER_12V_SENSE
Use Pre-Filter	Pre-Filter	Pre-Filter	Pre-Filter
NO Pre-Filter	Voltage_Monitor	Voltage_Monitor	N/A

Signal	Direction	Function
3V3	INPUT	Sense the 3V3 Voltage from PCIe golden finger
12V	INPUT	Sense the 12V Voltage from PCIe golden finger
PS_VMON_PGOOD	OPEN-DRAIN	Floating(H) once both 3V3 and 12V reach Vth
GC6_FB_EN	INPUT	Indicator for RTD3/GC6 residence.Use to Mask the VMON_PGOOD
PS_FF_SKIP	INPUT	From INA3221(VPU) or Pre-filter(SKIP)
PS_FF_BSCOK	INPUT	From INA3221(PV) or Pre-filter(BS_OK)

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Custom	0050_SEQ_VOLTAGE MONITOR	2.0
Date:	Friday, July 31, 2020	Sheet 50 of 55

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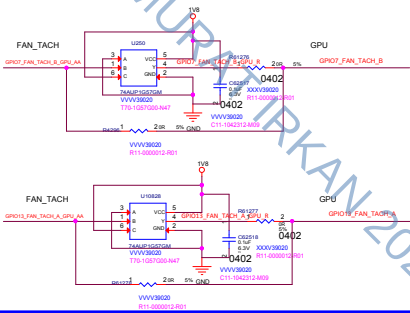
NV_PN	600-1G142-BASE-100
PCB REV	PG142-B00
BOM REV	A

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2-PIN LED HEADER
remove 2-PIN LED HEADER

copy from V388-2.0



footprint:TOWS_SIP4_2MM_1

J15在J16的左邊,白色接口。

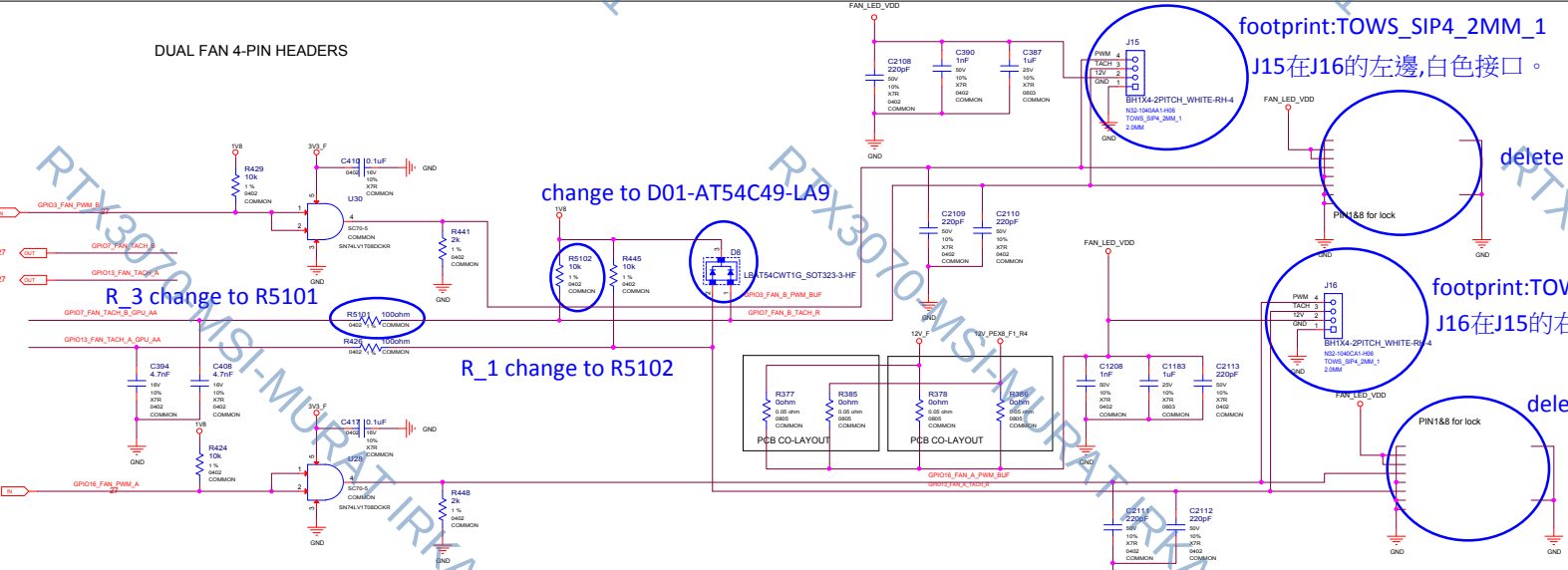
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footprint:TOWS_SIP4_2MM_1

J16在J15的右邊,黑色接口。

delete J501

DUAL FAN 4-PIN HEADERS



change to D01-AT54C49-LA9

R_3 change to R5101

R_1 change to R5102

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MS-V390			
Size	Custom	Document Description	Rev
Date:	Fri, July 31, 2020	0051_FAN & LED	2.0
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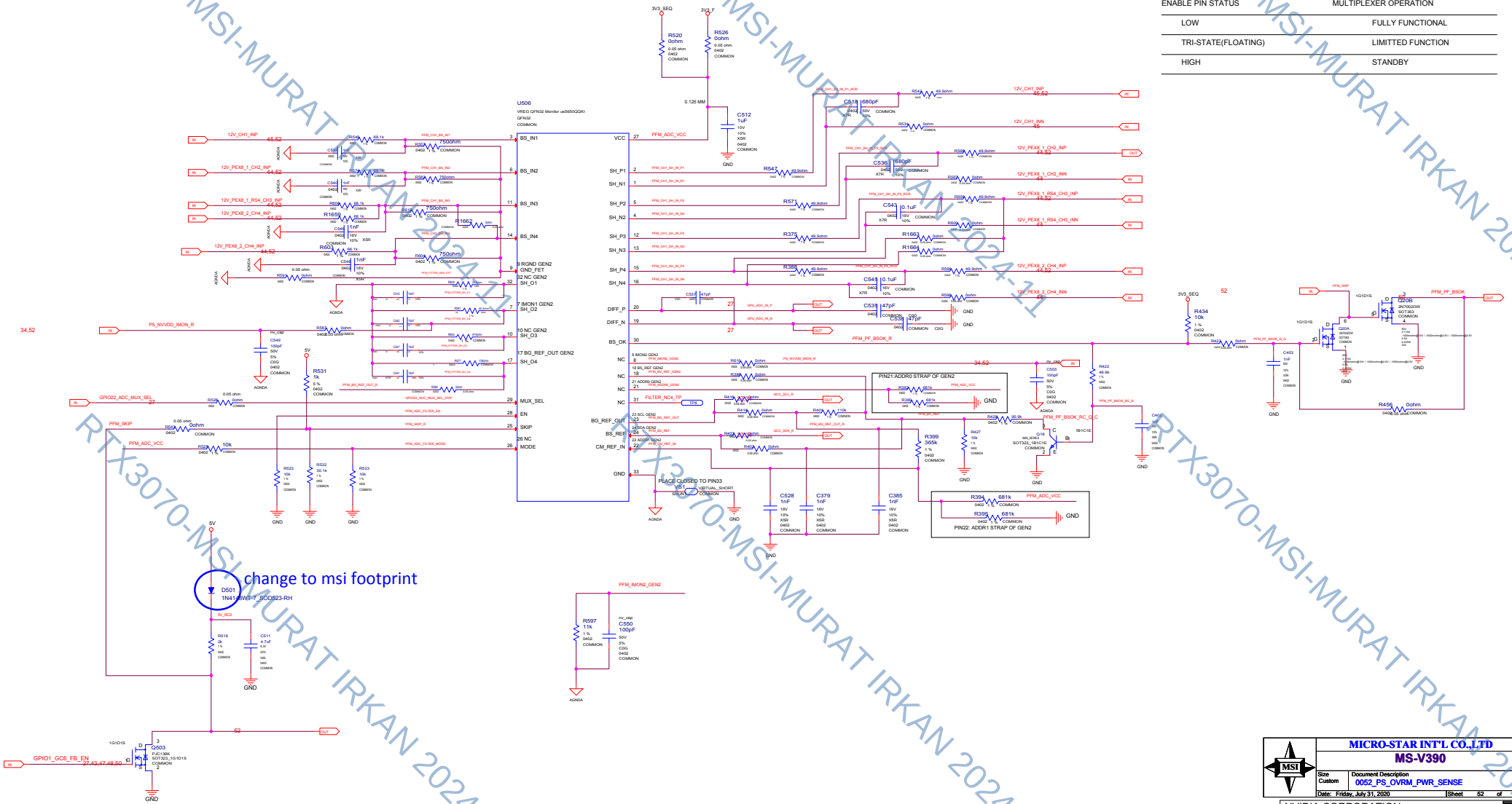
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MODE PIN STATUS		MULTIPLEXER OPERATION	
LOW		DEVICE A	
TRI-STATE(FLOATING)		STAND-ALONE	
HIGH		DEVICE B	

ENABLE PIN STATUS		MULTIPLEXER OPERATION	
LOW		FULLY FUNCTIONAL	
TRI-STATE(FLOATING)		LIMITED FUNCTION	
HIGH		STANDBY	





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Size

Custom

Document Description

0052_PS_OVRM_PWR_SENSE

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600-1G142-BASE-100

PCB REV

A

DATE

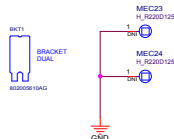
27-JUN-2020

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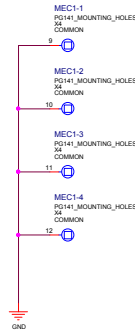
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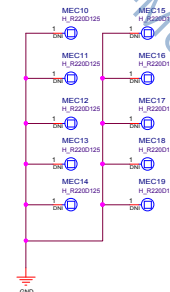
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footprint:H_R220D125



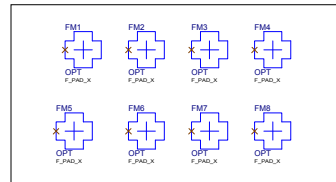
Fan module hole: 4 pcs
footprint : same as public board



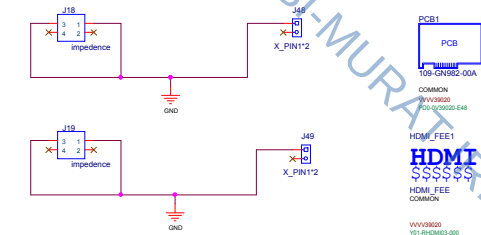
Fan module hole: 10 pcs
footprint:H_R220D125



删除CLP1~CLP3



Bracket Screw



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Custom	0053_MECH		
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